

Carrier Library

A complete reference of formal systems as $P/G \rightarrow Q$

Every formal system ever built is a specific choice of three parameters: a value space V , a gradient family G , and a coherence threshold θ . The propagation operator $P/G \rightarrow Q$ is constant across all of them. This catalogue presents 54 carriers across 7 sections, with forced laws, known failure modes, PL insights, and practical uses for each.

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$P / G \rightarrow Q$ with V, G, θ as the only free parameters

The mechanism is constant. The carrier is chosen. The rules follow.

Propositional Logic

14 carriers

01 Classical Logic

Propositional Logic

Aristotle (384 BC); formalised Frege (1879), Whitehead & Russell (1910)

V $\{0, 1\}$ -- exactly two truth values, no others

G $\text{NOT}(v) = 1-v$. $\text{AND} = \text{product}$. $\text{OR} = \text{max}$. $\text{IMP}(a,b) = \text{max}(1-a,b)$. NAND , NOR derived.

0 1.0 -- only full truth ($v=1$) is accepted

PL INSIGHT

Requiring every proposition to take exactly one of two values and demanding that NOT be an involution with no fixed point forces $V = \{0,1\}$ uniquely. LNC ($v*(1-v)=0$) and LEM ($\text{max}(v,1-v)=1$) then follow by arithmetic -- they are consequences of the carrier structure, not additional axioms. The binary carrier forces complete commitment on every proposition and cannot represent gradations without information loss.

Forces

+ LNC: $\text{AND}(v, \text{NOT}(v)) = v*(1-v) = 0$ for all v in $\{0,1\}$

+ LEM: $\text{OR}(v, \text{NOT}(v)) = \text{max}(v,1-v) = 1$ for all v in $\{0,1\}$

+ Double negation: $\text{NOT}(\text{NOT}(v)) = 1-(1-v) = v$

+ Ex falso: $\text{AND}(v, \text{NOT}(v)) = 0 \rightarrow \text{OR}(0, Q) = Q$ for any Q

Breaks on

✗ No fixed point for NOT: no v in $\{0,1\}$ satisfies $\text{NOT}(v)=v$ -- Liar paradox is unprovable

✗ No gradations: $v=0.5$ cannot be represented -- forces bivalence

Useful for

All classical mathematics and formal proofs

Digital circuit design

SAT solvers and theorem provers

Database query logic (SQL)

02 Lukasiewicz Three-Valued Logic (L3)

Propositional Logic

Jan Lukasiewicz (1920) -- O logice trojwartosciowej

V $\{0, 1/2, 1\}$ -- false, indeterminate, true

G $\text{NOT}(v) = 1-v$. $\text{AND}(a,b) = \text{max}(0, a+b-1)$. $\text{OR}(a,b) = \text{min}(1, a+b)$. $\text{IMP}(a,b) = \text{min}(1, 1-a+b)$.

0 1.0 -- only 1 is designated (fully accepted)

PL INSIGHT

Adding a midpoint value $1/2$ with $\text{NOT}(1/2) = 1-1/2 = 1/2$ creates a fixed point of negation -- the first value that is its own negation. This fixed point resolves the Liar sentence: it settles at $1/2$ rather than oscillating between 0 and 1. The strong t-norm $\text{AND}(a,b) = \text{max}(0,a+b-1)$ requires $a+b > 1$ to exceed 0, so two half-truths produce no truth at all. LNC holds ($\text{max}(0,0)=0$) and LEM holds ($\text{min}(1,1)=1$) at every point.

Forces

+ LNC: $\text{max}(0, v+(1-v)-1) = \text{max}(0,0) = 0$ for all v -- holds

+ LEM: $\text{min}(1, v+(1-v)) = \text{min}(1,1) = 1$ for all v -- holds

+ Fixed point: $\text{NOT}(1/2) = 1-1/2 = 1/2$ -- Liar resolves stably at $1/2$

Breaks on

✗ LEM fails for designated interpretation if $1/2$ is designated too

✗ $\text{AND}(1/2, 1/2) = \text{max}(0, 0) = 0$ -- two half-truths produce no truth

Useful for

Paracomplete logic (truth gaps)

Self-referential language analysis

Three-valued database NULL semantics (partial)

03 Kleene Strong Three-Valued Logic (K3)

Propositional Logic

Stephen Kleene (1952) -- *Introduction to Metamathematics*, pp. 332-340

V $\{0, 1/2, 1\}$ -- false, undefined, true
G $\text{NOT}(v) = 1-v$. $\text{AND}(a,b) = \min(a,b)$. $\text{OR}(a,b) = \max(a,b)$. $\text{IMP}(a,b) = \max(1-a,b)$.
0 1.0 -- only $v=1$ is designated

PL INSIGHT

Using the same three values as L3 but with $\text{AND}=\min$ and $\text{OR}=\max$, the intermediate value $1/2$ becomes a propagating taint rather than a stable midpoint. $\min(1/2, \text{anything})$ is at most $1/2$ and $\max(1/2, \text{anything})$ is at least $1/2$, so $1/2$ contaminates every conjunction below full truth and every disjunction above falsehood. Both LNC and LEM fail: $\text{AND}(1/2, \text{NOT}(1/2)) = \min(1/2, 1/2) = 1/2$ and $\text{OR}(1/2, \text{NOT}(1/2)) = \max(1/2, 1/2) = 1/2$. The value $1/2$ models a computation that has not yet returned -- its undefined status spreads through every operation that touches it.

Forces

- + AND and OR satisfy all lattice laws (associativity, commutativity, distributivity)
- + Double negation holds: $\text{NOT}(\text{NOT}(v)) = v$
- + Kleene's strong tables: identical to L3 for $\{0,1\}$ inputs

Breaks on

- × LNC: $\text{AND}(1/2, \text{NOT}(1/2)) = \min(1/2, 1/2) = 1/2$ not 0
- × LEM: $\text{OR}(1/2, \text{NOT}(1/2)) = \max(1/2, 1/2) = 1/2$ not 1

Useful for

Partial function semantics Three-valued database NULL (SQL weak semantics) Partial evaluation in compilers

04 Kleene Weak Three-Valued Logic

Propositional Logic

Stephen Kleene (1952) -- *weak tables variant*

V $\{0, u, 1\}$ -- false, undefined (u), true
G $\text{NOT}(u) = u$. $\text{AND}(u,x) = u$ and $\text{OR}(u,x) = u$ for all x -- u absorbs unconditionally. AND and OR on $\{0,1\}$ act classically.
0 1.0 -- only $v=1$ designated; u blocks all inference through it

PL INSIGHT

Weak K3 bans short-circuit evaluation by making u absorb without exception: $\text{AND}(u,x)=u$ and $\text{OR}(u,x)=u$ for all x , including $x=0$ and $x=1$. Strong K3 allows $\min(0,u)=0$ (known falsehood wins) and $\max(1,u)=1$ (known truth wins); Weak K3 prohibits both. The gradient family forbids the known value of one argument from rescuing a computation where the other argument is undefined. Every subexpression must resolve before any result can be determined -- strict call-by-value semantics encoded in the operator definitions.

Forces

- + u absorbs in both AND and OR regardless of the other argument
- + Restricts to classical logic on $\{0,1\}$ inputs

Breaks on

- × $\text{AND}(u, 0) = u$ (Strong K3 gives $\min(u,0) = 0$ -- known falsehood can win there)
- × $\text{OR}(u, 1) = u$ (Strong K3 gives $\max(u,1) = 1$ -- known truth can win there)
- × LNC and LEM both fail at u

Useful for

Strict evaluation semantics in programming languages Call-by-value lambda calculus models Error propagation analysis

05 Bochvar External Three-Valued Logic (B3)

Propositional Logic

Dmitri Bochvar (1938) -- *On a three-valued logical calculus and its application to paradoxes*

V $\{0, m, 1\}$ -- false, meaningless/paradoxical (m), true
G Internal: $\text{AND}(m, x) = m$, $\text{OR}(m, x) = m$ -- m absorbs. External assertion E : $E(1) = 1$, $E(0) = E(m) = 0$.
0 1.0 externally; internal reasoning quarantines m before reaching classical conclusions

PL INSIGHT

B3 uses a two-tier operator structure. Internally, the value m is absorbing: any formula containing m evaluates to m regardless of other arguments -- paradoxes are quarantined rather than resolved or eliminated. A second tier of external operators then collapses m to 0 before it can contaminate classical reasoning: $E(1) = 1$, $E(0) = 0$, $E(m) = 0$. The two tiers partition reasoning cleanly: the internal tier handles pathological inputs by containment, and the external tier restores classical decidability for sound formulas.

Forces

- + m is absorbing: any formula touching m internally evaluates to m
- + External E collapses m to 0, preserving classical decidability for well-formed formulas

Breaks on

- × No internal LNC or LEM for m -valued formulas
- × Classical rules inapplicable inside the m -tier

Useful for

Paradox containment in formal systems Partial information reasoning Undefined-value handling in formal semantics

06 Paraconsistent Logic LP (Priest)

Propositional Logic

Graham Priest (1979) -- *The Logic of Paradox, Journal of Philosophical Logic*

V $\{0, B, 1\}$ -- false, both (contradictory), true. $B = 1/2$ arithmetically.
G $\text{NOT}(v) = 1 - v$. $\text{AND} = \min$. $\text{OR} = \max$. B is designated (inferences from B -valued premises are acceptable).
0 Designated values: $\{B, 1\}$ -- both contradictory and true propositions license inference

PL INSIGHT

LP makes B a fixed point of negation: $\text{NOT}(B) = 1 - 1/2 = 1/2 = B$, so B is self-negating. The critical mechanism is $\text{AND} = \min$: $\text{AND}(B, \text{NOT}(B)) = \min(1/2, 1/2) = 1/2 = B$, not 0. LNC fails, but contradiction does not explode because the min operation cannot amplify B above itself to produce arbitrary conclusions. B is designated, so contradictory propositions remain usable for inference. Ex falso quodlibet is blocked precisely because the ceiling imposed by min prevents any chain of AND operations from raising a B -valued seed above B .

Forces

- + $\text{NOT}(B) = 1 - 1/2 = 1/2 = B$ -- contradiction is self-negating
- + $\text{AND}(B, \text{NOT}(B)) = \min(1/2, 1/2) = 1/2 = B$ -- LNC fails
- + Ex falso blocked: min cannot amplify B above B

Breaks on

- × LNC: $\text{AND}(B, \text{NOT}(B)) = B$ not 0
- × Classical explosion blocked but some paraconsistent-valid inferences differ from classical

Useful for

Formal dialethism (true contradictions) Inconsistent database reasoning Naive set theory without explosion

07 Fuzzy Logic (Zadeh / Product t-norm)

Propositional Logic

Lotfi Zadeh (1965) -- *Fuzzy Sets, Information and Control* 8(3)

V $[0, 1]$ -- continuous degrees of truth

G $\text{NOT}(v) = 1-v$. $\text{AND}(a,b) = a*b$ (product). $\text{OR}(a,b) = \max(a,b)$. $\text{IMP}(a,b) = b/a$ if $a>b$ else 1 (Goguen).

0 1.0 -- only $v=1$ is fully true; interior values are partial

PL INSIGHT

Extending V from $\{0,1\}$ to $[0,1]$ makes AND and OR continuous operators. Product AND ($a*b$) always gives a result weaker than either input in the interior: $\text{AND}(1/2, 1/2) = 1/4$, not $1/2$. OR = max takes the stronger. LNC fails because $v*(1-v) = 0.25$ at $v=0.5$: partial truth and partial falsehood produce non-zero conjunction. LEM fails because $\max(0.5, 0.5) = 0.5$, not 1. The Goguen implication (b/a when $a>b$) makes implication a ratio of strengths rather than a binary step.

Forces

+ Product is monotone: $a*b \leq \min(a,b)$ -- AND weakens

+ OR=max is associative, commutative, idempotent

+ $\text{IMP}(a,a) = 1$ for all $a>0$

Breaks on

✗ LNC: $v*(1-v) = 0.25$ at $v=0.5$ not 0

✗ LEM: $\max(0.5, 0.5) = 0.5$ not 1

✗ IMP undefined when $a=0$ (requires convention $\text{IMP}(0,b)=1$)

Useful for

Fuzzy control systems (washing machines, ABS brakes)

Approximate reasoning in AI

Soft constraints in optimization

08 Fuzzy Logic (Lukasiewicz / MV-algebra)

Propositional Logic

Lukasiewicz infinite-valued extension (1930); MV-algebra: Chang (1958)

V $[0, 1]$ -- continuous, metric completion of L_n as $n \rightarrow \infty$

G $\text{NOT}(v) = 1-v$. $\text{AND}(a,b) = \max(0, a+b-1)$. $\text{OR}(a,b) = \min(1, a+b)$. $\text{IMP}(a,b) = \min(1, 1-a+b)$.

0 1.0 -- only $v=1$ is designated

PL INSIGHT

The Lukasiewicz t-norm $\text{AND}(a,b) = \max(0, a+b-1)$ is the unique continuous extension of classical AND to $[0,1]$ that simultaneously preserves both LNC and LEM everywhere. LNC: $\max(0, v+(1-v)-1) = \max(0,0) = 0$ for all v . LEM: $\min(1, v+(1-v)) = \min(1,1) = 1$ for all v . No other continuous t-norm achieves both simultaneously. This algebra is also the metric completion of the sequence of finite L_n systems -- taking the limit as n grows without bound yields exactly this t-norm.

Forces

+ LNC: $\max(0, v+(1-v)-1) = 0$ for all v in $[0,1]$

+ LEM: $\min(1, v+(1-v)) = 1$ for all v in $[0,1]$

+ t-norm is the unique continuous one preserving both LNC and LEM

Breaks on

✗ Associativity holds only for the Lukasiewicz t-norm -- distinguishes it from product

Useful for

Algebraic logic (MV-algebras)

Formal fuzzy arithmetic

Probability logic connections

09 Fuzzy Logic (Godel / Min t-norm)

Propositional Logic

Kurt Godel (1932) -- Zum intuitionistischen Aussagenkalkul; Dummet (1959) LC

V $[0, 1]$ -- linearly ordered; IMP crisp
G $\text{NOT}(v) = 0$ if $v > 0$ else 1. $\text{AND}(a, b) = \min(a, b)$. $\text{OR}(a, b) = \max(a, b)$. $\text{IMP}(a, b) = 1$ if $a \leq b$ else b .
0 1.0 -- threshold for designated values

PL INSIGHT

Godel negation jumps discontinuously: $\text{NOT}(v)=0$ for all $v > 0$, and $\text{NOT}(0)=1$. This erases all gradient information under negation -- values 0.01 and 0.99 both negate to 0. $\text{AND}=\min$ preserves the weakest conjunct; $\text{OR}=\max$ preserves the strongest. Implication is crisp: $\text{IMP}(a, b)$ is always exactly 1 or b , never intermediate. This creates a logic where conjunction and disjunction are graded but negation and implication are binary steps, suited to comparison reasoning without symmetric treatment of the two poles.

Forces

- + $\text{AND}=\min$ is idempotent: $\min(a, a)=a$
- + IMP is always in $\{0, 1\}$ or $\{b\}$: implication is crisp
- + Lattice laws for min/max hold throughout $[0, 1]$

Breaks on

- × NOT is discontinuous: any $v > 0$ negates to 0
- × Double negation fails: $\text{NOT}(\text{NOT}(v)) = 0$ for any $v > 0$
- × LEM fails: $\max(v, 0) = v$ not 1 for v in $(0, 1)$

Useful for

Intermediate logics between intuitionistic and classical Graded comparison and ordering Formal semantics of 'at least as true as'

10 Intuitionistic Logic (BHK semantics)

Propositional Logic

Brouwer (1908); Heyting (1930); Kolmogorov (1932) -- BHK interpretation

V $\{0, 1\}$ as values, but only constructible by explicit proof; no oracle for LEM
G AND-intro requires proofs of both conjuncts. OR-intro requires a proof of one disjunct plus a witness. IMP-intro requires a function from proofs of P to proofs of Q . $\text{NOT}(P) = P \rightarrow \text{false}$.
0 1.0 -- a proposition is true only when a proof has been explicitly constructed

PL INSIGHT

Intuitionistic logic keeps $V = \{0, 1\}$ but changes the disjunction rule: to prove $P \text{ OR } Q$ you must produce a proof of P or a proof of Q -- asserting the disjunction without a witness for one disjunct is not a valid operation. LEM ($P \text{ OR } \text{NOT-}P$) is arithmetically sound but constructively unprovable for any undecided proposition: neither a proof of P nor a proof of $\text{NOT-}P$ may be available. Double negation fails for the same reason: $\text{NOT-}P$ says refuting P is impossible, but impossibility of refutation does not construct a proof.

Forces

- + LNC: a proof of P and of $\text{NOT-}P$ yields a proof of false (ex falso valid)
- + AND-intro, OR-intro (with witness), IMP-intro, IMP-elim: all valid
- + Double negation introduction: $P \rightarrow \text{NOT-}P$ (always valid)

Breaks on

- × LEM: no proof of $P \text{ OR } \text{NOT-}P$ without a prior proof of P or of $\text{NOT-}P$
- × Double negation elimination: $\text{NOT-}P \rightarrow P$ fails (knowing refutation is impossible does not construct P)
- × $\text{IMP}(A, B \text{ OR } C)$ does not force $\text{IMP}(A, B)$ OR $\text{IMP}(A, C)$ in general

Useful for

Constructive mathematics (Bishop, Martin-Lof) Proof assistants (Coq, Agda, Lean) Type theory (Curry-Howard: proofs as programs)

11 Linear Logic

Propositional Logic

Jean-Yves Girard (1987) -- *Linear Logic, Theoretical Computer Science 50(1)*

- V** Proof-nets / sequents; formulas as resources used exactly once (without !)
- G** Tensor AND (A tensor B): both resources simultaneously. Additive AND (A & B): choose one. ! A: unlimited supply. Linear IMP (A -o B): consuming A produces B.
- 0** Exact resource count; each formula consumed exactly once unless marked !

PL INSIGHT

Linear logic tracks resource consumption structurally: every formula is used exactly once unless annotated with ! (of-course, meaning 'unlimited supply'). Tensor AND requires both resources to be produced and consumed. Additive AND offers a choice -- one resource is consumed, the other discarded. Without !, Weakening (introducing unused assumptions) and Contraction (using an assumption twice) are invalid proof steps. The result is a proof calculus where every hypothesis is fully accounted for, corresponding to physical resource conservation: every use has a cost, every cost is tracked.

Forces

- + Modus ponens with linear IMP: A tensor (A -o B) |- B (resource transferred)
- + Tensor associativity and commutativity (with exchange rule)
- + ! A restores classical structural rules (weakening + contraction) for marked formulas

Breaks on

- × Contraction: A |- A tensor A is invalid (cannot copy a resource)
- × Weakening: A, B |- A is invalid (cannot discard a purchased resource)
- × Classical explosion: A tensor NOT-A |- B does not hold without !

Useful for

- Type theory for resource-aware computation
- Rust ownership model (formal semantics)
- Session types in concurrent programming
- Proof complexity lower bounds

12 Relevance Logic (R, E)

Propositional Logic

Wilhelm Ackermann (1956); Anderson & Belnap (1975) -- *Entailment Vol. I*

- V** {0, 1} or Routley-Meyer semantics; relevance filter on proof steps
- G** All classical connectives preserved, but IMP(A,B) valid only when A and B share a propositional variable. Proof graphs must be connected.
- 0** Sharing requirement: every premise must contribute a variable to the conclusion

PL INSIGHT

Relevance logic adds a variable-sharing constraint to implication: $P \rightarrow Q$ is derivable only when P and Q share at least one propositional variable. This eliminates paradoxes of material implication where classical logic validates $P \rightarrow (Q \rightarrow P)$ -- adding any irrelevant antecedent -- or $\text{NOT-}P \rightarrow (P \rightarrow Q)$ -- anything follows from a false premise. The constraint ensures every step in a derivation actually uses the content of the cited premise; inference must be semantically relevant, not merely logically compatible.

Forces

- + Modus ponens (A shares variables with $A \rightarrow B$ by definition)
- + AND-intro, OR-intro: both valid
- + Self-implication: $A \rightarrow A$ always valid (trivially shares)

Breaks on

- × Paradoxes of material implication: $P \rightarrow (Q \rightarrow P)$ invalid (P and Q may share nothing)
- × Ex falso: $\text{NOT-}P, P \rightarrow Q$ invalid if P and Q are unrelated
- × Permutation: $(A \rightarrow (B \rightarrow C)) \rightarrow (B \rightarrow (A \rightarrow C))$ requires shared variables

Useful for

- Conditional logic and relevant implication
- Adaptive logics (belief revision)
- Formal models of scientific explanation

13 Dual-Intuitionistic Logic (co-Heyting)

Propositional Logic

Rauszer (1974); Goodman (1981) -- categorical dual of intuitionistic logic

- ✓ Co-Heyting algebras; truth values dual to open sets (closed sets of a topological space)
- ✓ Subtraction $A - B$ (co-implication / co-Heyting): the least A such that $B \leq A \vee C$. $\vee$ requires explicit refutation evidence rather than proof.
- ⊖ 0.0 -- a proposition fails only when a refutation has been explicitly constructed

PL INSIGHT

Dual-intuitionistic logic takes the categorical dual of every intuitionistic operation: subtraction $A-B$ (co-implication) replaces implication as the primitive connective, and $\vee$ requires explicit refutation evidence rather than a proof witness. Where constructive logic asks 'do you have a proof of P ?', co-intuitionistic logic asks 'do you have a refutation of $\text{NOT-}P$?'. Refutations are the primitive certificates. The system is suited to default reasoning: propositions are accepted as-is until a specific refutation is produced against them.

Forces

- + Co-LNC (dual of LEM): $A \vee \text{NOT-}A$ valid (open sets cover; dual: a closed set and its boundary)
- + Subtraction $A - B$ exists as the smallest set containing A but not B
- + De Morgan duals of all intuitionistic theorems

Breaks on

- ✗ Co-LEM (dual of LNC): $A \wedge \text{NOT-}A$ fails in general
- ✗ Classical explosion: $A \wedge \text{NOT-}A \rightarrow B$ does not hold without additional structure
- ✗ IMP-intro is not a primitive operation

Useful for

Default reasoning and defeasible inference Paracomplete logic (dual) Abstract interpretation (over-approximation duals)

14 Substructural Logic (general framework)

Propositional Logic

Gentzen (1935) structural rules; Došen (1988) substructural; Restall (2000) comprehensive

- ✓ Sequents $\Gamma \vdash \Delta$; $\vdash$ depends on retained structural rules
- ✓ Connectives identical to classical logic; structural rules are the variable. Weakening (W), Contraction (C), Exchange (E), Cut each independently removable.
- ⊖ Depends on which structural rules are kept; removing rules restricts valid proofs

PL INSIGHT

Classical logic silently assumes four structural rules: Weakening (introduce unused hypotheses), Contraction (reuse a hypothesis), Exchange (reorder premises), and Cut (chain two proofs via an intermediate). Substructural logics make these rules explicit and remove them selectively. Linear logic removes Contraction: each premise is used exactly once. Relevance logic removes Weakening: every hypothesis must be used. The Lambek calculus removes Exchange: the order of premises is significant. Removing each rule corresponds to a real constraint -- on resources, on relevance, on sequentiality -- that classical logic ignores.

Forces

- + Identity: $A \vdash A$ always valid
- + Cut elimination (in most substructural logics)
- + Remaining structural rules force their associated theorems

Breaks on

- ✗ Removing W : weakening -- hypotheses must be used
- ✗ Removing C : contraction -- hypotheses used at most once
- ✗ Removing E : exchange -- order of premises matters (Lambek calculus)
- ✗ Removing all: minimal logic

Useful for

Type theory foundations Categorical grammars (Lambek calculus models natural language syntax) Proof complexity
Resource-bounded computation

15 Modal Logic K (Kripke base)

Modal Logic

C.I. Lewis (1918) modal axioms; Saul Kripke (1959-1963) possible worlds semantics

- V** Set of worlds W ; each world w assigns truth values to propositions. $\text{NECS}(P)$ and $\text{POSS}(P)$ index truth over accessible worlds.
- G** $\text{NECS}(P)$ at w : P holds at all w' with wRw' (accessibility). $\text{POSS}(P)$ at w : P holds at some w' with wRw' . No constraint on R .
- Ø** Accessibility relation R ; more constrained R = stronger modal logic

PL INSIGHT

Modal logic K indexes truth at worlds in a set W with a binary accessibility relation R between worlds. $\text{NECS}(P)$ at world w requires P to hold at every world w' with wRw' : a universal quantification over R -neighbors. $\text{POSS}(P)$ requires P at some accessible w' : existential quantification. No constraint is placed on R , so worlds may not reach themselves (no T axiom), chains may not close (no 4 axiom), and accessibility may be asymmetric. Each additional constraint on R generates a strictly stronger modal logic.

Forces

- + NECS distributes over AND: $\text{NECS}(P \text{ AND } Q) \leftrightarrow \text{NECS}(P) \text{ AND } \text{NECS}(Q)$
- + NECS and POSS are duals: $\text{POSS}(P) = \text{NOT}(\text{NECS}(\text{NOT}(P)))$
- + K axiom: $\text{NECS}(P \rightarrow Q) \rightarrow (\text{NECS}(P) \rightarrow \text{NECS}(Q))$ (modal modus ponens)

Breaks on

- × T axiom $\text{NECS}(P) \rightarrow P$: not forced (worlds may not see themselves)
- × 4 axiom $\text{NECS}(P) \rightarrow \text{NECS}(\text{NECS}(P))$: not forced (transitivity not assumed)
- × 5 axiom $\text{POSS}(P) \rightarrow \text{NECS}(\text{POSS}(P))$: not forced (symmetry not assumed)

Useful for

- Temporal logic (accessibility = time steps)
- Deontic logic (accessibility = ideal worlds)
- Provability logic (accessibility = proof steps)
- Knowledge representation

16 Modal Logic T (Reflexive frames)

Modal Logic

Feys (1937); von Wright (1951) -- adds reflexivity to K

- V** Kripke frames where every world sees itself: wRw for all w in W
- G** All K gradients plus: wRw forces $\text{NECS}(P)$ at w to include the current world in its universal quantification.
- Ø** Reflexive accessibility: worlds are self-accessible

PL INSIGHT

Making the accessibility relation reflexive -- each world sees itself, wRw for all w -- forces the axiom $\text{NECS}(P) \rightarrow P$ directly. The universal quantification over accessible worlds now includes the current world, so if P holds at all accessible worlds, P holds here. No additional semantic machinery is required: reflexivity alone forces necessity to imply actuality. The T axiom is not an added postulate but a direct consequence of the reflexivity constraint on R .

Forces

- + T axiom: $\text{NECS}(P) \rightarrow P$ (forced by wRw -- P holds at all accessible worlds, including w)
- + $\text{POSS}(P) \rightarrow P$ not forced (P might hold only at w , not at other accessible worlds)

Breaks on

- × 4 axiom: $\text{NECS}(P) \rightarrow \text{NECS}(\text{NECS}(P))$ not forced without transitivity
- × B axiom not forced without symmetry

Useful for

- Alethic modality (necessary truth)
- Epistemic logic with truthfulness assumption
- Base for S4 and S5 systems

17 Modal Logic S4 (Reflexive + Transitive)

Modal Logic

Lewis & Langford (1932); Kripke semantics: reflexive + transitive frames

- V** Kripke frames where R is a preorder (reflexive and transitive)
- G** K plus T (reflexivity) plus 4 (transitivity): if wRv and vRu then wRu .
- Θ** Transitive accessible worlds; knowing that you know

PL INSIGHT

Transitivity of accessibility (wRv and vRu implies wRu) forces the 4-axiom $\text{NECS}(P) \rightarrow \text{NECS}(\text{NECS}(P))$. If P holds in all worlds accessible from w, and all worlds accessible from those worlds are also accessible from w by transitivity, then 'P holds in all accessible worlds from w' is itself a fact about all worlds accessible from w -- making it necessary. Combined with reflexivity (T axiom), the system captures positive introspection: what is necessary is necessarily so, and the agent knows what it knows.

Forces

- + T: $\text{NECS}(P) \rightarrow P$ (reflexivity)
- + 4: $\text{NECS}(P) \rightarrow \text{NECS}(\text{NECS}(P))$ (transitivity -- worlds reachable from reachable worlds are reachable from here)
- + Positive introspection: what is necessary is necessarily necessary

Breaks on

- ✗ B axiom: $P \rightarrow \text{NECS}(\text{POSS}(P))$ not forced (symmetry absent)
- ✗ 5 axiom not forced -- an agent may not know what it does not know

Useful for

Epistemic logic (knowledge without knowing ignorance) Tense logic for future necessity Intuitionistic logic (Kripke semantics via S4 translation)

18 Modal Logic S5 (Equivalence Relation)

Modal Logic

Lewis (1932); Kripke semantics: equivalence relation on worlds

- V** Kripke frames where R is an equivalence relation (reflexive + transitive + symmetric)
- G** All S4 gradients plus symmetry: wRv implies vRw . Every world in an equivalence class sees every other.
- Θ** Full mutual accessibility within equivalence classes

PL INSIGHT

When the accessibility relation is an equivalence (reflexive + transitive + symmetric), the two modal operators take their simplest form: necessity is universal truth over the equivalence class, possibility is existential truth. Symmetry forces the 5-axiom: if P is possibly necessary from w, then from some accessible world P is necessary over all worlds in the class -- since those are the same worlds accessible from w, P is necessary here. Within each equivalence class, all worlds are interchangeable for modal purposes, and $\text{NECS}(P)$ with $\text{POSS}(\text{NOT-}P)$ become strict contradictions.

Forces

- + T and 4 from S4
- + 5: $\text{POSS}(P) \rightarrow \text{NECS}(\text{POSS}(P))$ (if P is possible from w, then from any world w can see, P is also possible -- since all worlds in the class see each other)
- + NECS and POSS become truth quantifiers over the equivalence class

Breaks on

- ✗ No further axioms to add in the standard system
- ✗ Cross-class inference not available

Useful for

Propositional logic (trivial accessibility = classical) Metaphysical modality Normal form for epistemic logics with full introspection

19 Provability Logic GL (Godel-Lob)

Modal Logic

Lob (1955); Solovay (1976) completeness; Boolos (1979) The Unprovability of Consistency

- V** Worlds = consistent extensions of PA; accessibility = provability in PA
- G** $\text{NECS}(P) = \text{'P is provable in PA'}$. Transitivity (derivability is transitive). Lob axiom: $\text{NECS}(\text{NECS}(P) \rightarrow P) \rightarrow \text{NECS}(P)$.
- Ø** Proof complexity depth; Godel sentences have exponential cost at depth d

PL INSIGHT

Godel-Lob logic uses $\text{NECS}(P)$ to mean 'P is provable in Peano Arithmetic'. Lob's theorem ($\text{NECS}(\text{NECS}(P) \rightarrow P) \rightarrow \text{NECS}(P)$) is an axiom: if PA proves 'provability of P implies P', then P is actually provable. The T-axiom $\text{NECS}(P) \rightarrow P$ fails -- PA cannot prove its own soundness (Godel's second incompleteness theorem). Self-referential sentences at reasoning depth d require simulating provability at depth $d-1$, doubling the verification work at each level, generating exponential complexity.

Forces

- + Lob's theorem: $\text{NECS}(\text{NECS}(P) \rightarrow P) \rightarrow \text{NECS}(P)$ (if PA proves 'provability implies truth', then P is provable)
- + 4 axiom: provability is transitive -- if PA proves P then PA proves 'PA proves P'
- + Well-foundedness: no infinite descending chains of provability

Breaks on

- × T axiom $\text{NECS}(P) \rightarrow P$ fails: PA cannot prove its own soundness (Godel incompleteness)
- × Self-referential sentences generate exponential verification complexity

Useful for

Formal analysis of Godel's incompleteness theorems Provability predicates in arithmetic Self-referential reasoning limits

20 Epistemic Logic (Multi-Agent)

Modal Logic

Hintikka (1962) -- Knowledge and Belief; Fagin, Halpern, Moses, Vardi (1995)

- V** Worlds in W ; each agent i has accessibility relation R_i . $K_i(P) = \text{agent } i \text{ knows } P$.
- G** $K_i(P)$ at w : P holds in all worlds agent i cannot distinguish from w . Common knowledge $C(P)$: fixed point of mutual knowledge operator.
- Ø** Epistemic indistinguishability per agent; common knowledge requires infinite closure

PL INSIGHT

Each agent i has a private accessibility relation R_i representing epistemic indistinguishability: $K_i(P)$ requires P to hold in every world agent i cannot distinguish from the current one. Common knowledge $C(P)$ is defined as the infinite conjunction: every agent knows P , and every agent knows that every agent knows P , and so on. This infinite nesting cannot be established by any finite sequence of communications -- each message round adds one layer of mutual knowledge but never reaches the fixed-point closure. Common knowledge requires simultaneous broadcast or an infinite process.

Forces

- + $K_i(P \text{ AND } Q) \leftrightarrow K_i(P) \text{ AND } K_i(Q)$ (knowledge distributes over AND)
- + $K_i(P) \rightarrow P$ (with truthfulness assumption, adding T axiom)
- + $K_i(P) \rightarrow K_i(K_i(P))$ (positive introspection, with S4 frames)

Breaks on

- × Common knowledge $C(P)$ is not achievable by finite communication: each round adds one layer of mutual knowledge but never reaches the fixed point
- × Negative introspection $\text{NOT}(K_i(P)) \rightarrow K_i(\text{NOT}(K_i(P)))$ requires S5 frames

Useful for

Distributed systems (consensus protocols) Game theory (common knowledge of rationality) Social epistemology
Cryptographic protocols (knowledge assumptions)

21 Temporal Logic (LTL / CTL)

Modal Logic

Prior (1957) tense logic; Pnueli (1977) LTL; Clarke & Emerson (1981) CTL

- V** States over time; LTL: linear sequences; CTL: branching computation trees
- G** $X(P)$: next step. $F(P)$: eventually (existential over future). $G(P)$: always (universal over all future). Until: $P \text{ U } Q = Q$ holds eventually, P holds until then.
- ⊖** Finite path length for F ; infinite for G (G is never verified by finite observation)

PL INSIGHT

Temporal logic treats time steps as worlds in a linear (LTL) or branching (CTL) accessibility structure. $X(P)$ asserts P holds at the immediately next step: one-step lookahead. $F(P)$ asserts P at some future step -- existential quantification over time with the earliest witness as the lower bound. $G(P)$ asserts P at every future step -- universal quantification over an infinite sequence that can never be verified by finite observation alone. Duality $F(P) = \text{NOT}(G(\text{NOT-}P))$ mirrors modal possibility and necessity. CTL branches over tree paths while LTL quantifies over a single linear future.

Forces

- + Duality: $F(P) = \text{NOT}(G(\text{NOT-}P))$ -- mirrors possibility/necessity
- + LTL: $G(P) \rightarrow P$ (current time is in 'all future steps' starting now)
- + $F(F(P)) \leftrightarrow F(P)$ -- idempotent: eventually-eventually = eventually

Breaks on

- × $G(P)$ undecidable by finite observation alone
- × CTL and LTL are incomparable: each can express properties the other cannot

Useful for

Hardware and software model checking (SPIN, NuSMV) Reactive system specification Program verification (liveness and safety properties)

22 Deontic Logic

Modal Logic

von Wright (1951) -- Deontic Logic; Chisholm (1963) paradox

- V** Worlds ranked by moral ideality; $O(P)$ = obligatory, $P(P)$ = permitted, $F(P)$ = forbidden
- G** $O(P)$: P holds in all ideal (morally optimal) accessible worlds. $P(P)$: P holds in some ideal world. $F(P) = O(\text{NOT-}P)$.
- ⊖** Moral ideality ordering on worlds; actual world may not be ideal

PL INSIGHT

Deontic logic designates certain worlds as morally ideal and uses the obligatory/permitted distinction to quantify over them: $O(P)$ when P holds in all ideal worlds accessible from the actual world; $P(P)$ when P holds in some. The structural laws follow from this setup. Contrary-to-duty paradoxes arise when the set of ideal worlds is internally inconsistent: $O(A)$ requires worlds where A holds, but $O(\text{NOT-}A \text{ given } A\text{-is-violated})$ points toward worlds incompatible with those, and no single world satisfies both ideality requirements simultaneously.

Forces

- + $O(P) \rightarrow P(P)$ (if obligatory, then permitted -- all ideal worlds is subset of some ideal worlds)
- + $O(P \text{ AND } Q) \leftrightarrow O(P) \text{ AND } O(Q)$ (distributivity over AND)
- + $\text{NOT}(P(P) \text{ AND } P(\text{NOT-}P))$ in some systems (excluded contradictory permission)

Breaks on

- × Contrary-to-duty paradoxes: $O(A)$ and $O(\text{NOT-}A \mid A\text{-violated})$ point to mutually inconsistent ideal worlds
- × Chisholm paradox: standard deontic axioms allow contradictory obligations when violations occur
- × P-axiom $P(T)$ (tautology is permitted) is added, not forced

Useful for

Formal ethics and AI alignment Legal reasoning and normative systems Multi-agent obligation specification

23 Post n-Valued Logic (P_n)

Many-Valued Logic

Emil Post (1921) -- *Introduction to a General Theory of Elementary Propositions*

V $\{0, 1/(n-1), 2/(n-1), \dots, 1\}$ -- n equally spaced values
G $\text{NOT}(k/(n-1)) = (k+1 \bmod n)/(n-1)$ -- cyclic shift of order n . AND = min. OR = max.
0 1.0 -- only full truth designated

PL INSIGHT

Post systems generalise classical negation from an involution (order 2) to a cyclic rotation of order n : NOT shifts each value one step along the cycle, and applying NOT n times returns to the original. AND=min and OR=max operate as in Kleene's K3. LNC and LEM both fail at interior values because NOT is no longer the complement $1-v$. The $n=2$ case recovers classical logic (the cycle has order 2 and exchanges 0 and 1). As n grows without bound, the equally spaced values become dense and the cyclic negation increasingly approximates the linear complement $1-v$.

Forces

- + Cyclic NOT has order n : applying NOT n times returns to start
- + AND/OR are lattice operations (same as K3-style)
- + $n=2$ reduces to classical logic (cycle of order 2: $0 \leftrightarrow 1$)

Breaks on

- ✗ LNC fails for interior values: $\min(v, \text{NOT}(v)) \neq 0$ when NOT is cyclic
- ✗ LEM fails for interior values: $\max(v, \text{NOT}(v)) \neq 1$ in general
- ✗ NOT is not the complement $1-v$ except at endpoints

Useful for

Finite-valued logic theory Error-correcting code design Many-valued switching circuit theory

24 Belnap Four-Valued Logic (FOUR)

Many-Valued Logic

Nuel Belnap (1977) -- *A useful four-valued logic; How a computer should think*

V $\{N, F, T, B\}$ -- None (no info), False (only false), True (only true), Both (conflict). Two independent orderings.
G Truth ordering: $F \leq_t N, B \leq_t T$. Knowledge ordering: $N \leq_k F, T \leq_k B$. AND and OR operate componentwise on each ordering.
0 T in the truth ordering is designated; B is the maximal information (contradictory) state

PL INSIGHT

Belnap's FOUR separates truth and information into two independent orderings on the same four values. The truth ordering ranks F below N and B below T . The knowledge ordering ranks N below F and T below B . N carries no information about either truth or falsity. B carries maximal information but contradictory: both true and false evidence has been received. These two orderings yield a bilattice structure where AND and OR operate independently on each dimension -- the dimensions that classical logic collapses into one are kept orthogonal.

Forces

- + Bilattice structure: two lattice orderings on the same four elements
- + $\text{NOT}(N)=N, \text{NOT}(F)=T, \text{NOT}(T)=F, \text{NOT}(B)=B$ -- NOT swaps truth poles, preserves information poles
- + N and B are both fixed under the knowledge ordering: they are the extremes of information

Breaks on

- ✗ Classical LNC fails: $\text{NOT}(B) = B$ (contradiction is self-negating)
- ✗ Classical LEM fails for N : N has no truth content to assert

Useful for

Inconsistent database management Paraconsistent reasoning over conflicting sources Multi-source information aggregation

25 MV-Algebra (Chang)

Many-Valued Logic

C.C. Chang (1958) -- *Algebraic analysis of many-valued logics*

- V** $[0, 1]$ or any MV-algebra; bounded domain with NOT and bounded addition as primitives
- G** $\text{PLUS}(a,b) = \min(1, a+b)$ -- bounded addition. $\text{NOT}(a) = 1-a$. $\text{AND} = \text{PLUS}(a, \text{NOT}(b)) \dots = \max(0, a+b-1)$. $\text{OR} = \text{NOT}(\text{PLUS}(\text{NOT}(a), \text{NOT}(b))) = \min(1, a+b)$.
- 0** 1.0; PLUS saturates at 1 preventing value overflow

PL INSIGHT

MV-algebras define their entire algebraic system from just two operations: bounded addition $\text{PLUS}(a,b) = \min(1, a+b)$ which saturates at 1, and complementation $\text{NOT}(a)=1-a$. All Lukasiewicz connectives are expressible in terms of these two: adding 0.7 and 0.8 gives 1, not 1.5. Chang's completeness theorem establishes that MV-algebra semantics is exactly complete for Lukasiewicz infinite-valued logic: a formula is derivable in the logic if and only if it evaluates to 1 in every MV-algebra. The two primitives generate the full system.

Forces

- + Chang completeness: a formula is Lukasiewicz-valid iff it evaluates to 1 in every MV-algebra
- + All Lukasiewicz connectives derivable from PLUS and NOT alone
- + PLUS is commutative, associative, and has identity 0

Breaks on

- × MV-algebras do not capture the product t-norm or Gödel t-norm
- × Infinite-valued MV-algebras are complete; finite ones correspond to L_n systems

Useful for

Algebraic semantics for Lukasiewicz logic Many-valued arithmetic Formal treatment of probability-like reasoning

26 Effect Algebra (Quantum Logic)

Many-Valued Logic

Foulis & Bennett (1994) -- *Effect algebras and unsharp quantum logics*

- V** Effects = quantum observables with outcome probabilities in $[0,1]$; partial algebraic structure
- G** Partial sum $x (+) y$ defined only when x is orthogonal to y ($x \perp y$). Orthocomplementation: $\text{NOT}(x) = x'$. Zero and unit elements.
- 0** Orthogonality: $x (+) y$ defined iff $x + y \leq 1$ in the $[0,1]$ interval (probability interpretation)

PL INSIGHT

An effect algebra formalizes which quantum measurements can be combined. The partial sum $x (+) y$ is defined only when x and y are orthogonal ($x \perp y$), meaning they represent simultaneously measurable quantities. When x and y are non-orthogonal -- like position and momentum -- $x (+) y$ is simply undefined: the operation does not exist at all, not merely equals zero. The Heisenberg uncertainty principle is a theorem of the algebra -- it follows from the undefined status of incompatible sums -- rather than a separately added postulate.

Forces

- + Commutativity: $x (+) y = y (+) x$ when defined
- + Associativity when defined
- + $x (+) \text{NOT}(x) = 1$ (every effect and its complement sum to certainty)
- + Cancellation: $x (+) y = x (+) z$ implies $y = z$

Breaks on

- × Non-orthogonal effects (e.g., position and momentum in QM): $x (+) y$ is undefined, not zero -- the operation does not exist
- × Classical Boolean algebra is a special case: all pairs orthogonal

Useful for

Quantum measurement theory Fuzzy quantum logic Operational quantum mechanics

27 Heyting Algebra

Many-Valued Logic

Arend Heyting (1930) -- algebraic semantics of intuitionistic logic

- V** Any Heyting algebra H : a lattice with implication as residual of meet
- G** MEET (AND): $a \text{ AND } b = \text{greatest lower bound}$. JOIN (OR): $a \text{ OR } b = \text{least upper bound}$. $\text{IMP}(a,b)$: largest c with $a \text{ AND } c \leq b$ (residual). $\text{NOT}(a) = \text{IMP}(a, 0)$.
- 0** Top element 1 (maximum); LEM fails when 1 is not reachable as a JOIN

PL INSIGHT

In a Heyting algebra, implication $a \rightarrow b$ is the largest element c satisfying $a \text{ AND } c \leq b$ -- the residual of meet. This gives the intuitionistic laws as algebraic consequences: LNC holds because $a \text{ AND } (a \rightarrow 0) \leq 0$ by the very definition of $\rightarrow$. LEM fails because $a \text{ OR } (a \rightarrow 0)$ need not reach the top element when the lattice has no complementation law forcing it. The open sets of any topological space form a Heyting algebra: truth at a region means truth on some neighbourhood, and a neighbourhood that refutes P need not be complemented by one that proves P .

Forces

- + Modus ponens: $a \text{ AND } \text{IMP}(a,b) \leq b$ (by definition of residual)
- + LNC: $a \text{ AND } \text{NOT}(a) = a \text{ AND } \text{IMP}(a,0) \leq 0$ (residual definition forces this)
- + $\text{IMP}(a,a) = 1$ for all a
- + Adjointness: $a \leq \text{IMP}(b,c)$ iff $a \text{ AND } b \leq c$

Breaks on

- ✗ LEM: $a \text{ OR } \text{NOT}(a)$ need not equal 1 in general Heyting algebras
- ✗ Double negation: $\text{NOT}(\text{NOT}(a)) = a$ fails -- $\text{NOT}(\text{NOT}(a)) \geq a$ but equality is not forced

Useful for

Algebraic semantics of intuitionistic logic Topos theory (subobject classifiers) Formal topology and locale theory Sheaf semantics

28 Basic Logic (BL) / Hajek

Many-Valued Logic

Petr Hajek (1998) -- Metamathematics of Fuzzy Logic

- V** $[0, 1]$ or any BL-algebra (MTL-algebra + divisibility condition)
- G** $*$ (any continuous t-norm as AND). $\rightarrow$ (its residuum). $\text{AND} = \min$. $\text{OR} = \max$. $\text{NOT}(a) = a \rightarrow 0$.
- 0** 1.0

PL INSIGHT

Basic Logic axiomatizes only the properties common to all continuous t-norm based systems. The requirements are: the t-norm $*$ is continuous with residuum $\rightarrow$ ($a*b \leq c$ iff $a \leq b \rightarrow c$), divisibility holds ($a \text{ AND } b = a*(a \rightarrow b)$), and prelinearity ($(a \rightarrow b) \text{ OR } (b \rightarrow a) = 1$). Every continuous t-norm generates a BL-algebra. The Lukasiewicz, Godel, and product t-norms are the three irreducible cases: every other continuous t-norm is an ordinal sum -- a piecewise combination of these three on intervals. BL captures the structural laws that hold regardless of which continuous conjunction is chosen.

Forces

- + Residuation: $a*b \leq c$ iff $a \leq b \rightarrow c$
- + Divisibility: $a \text{ AND } b = a*(a \rightarrow b)$
- + Prelinearity: $(a \rightarrow b) \text{ OR } (b \rightarrow a) = 1$

Breaks on

- ✗ LEM not forced (intermediate values)
- ✗ LNC fails for non-product t-norms at interior points

Useful for

Fuzzy logic unification Truth-functional many-valued reasoning Formal fuzzy arithmetic

Set Theory & Type Theory

7 carriers

29 Zermelo-Fraenkel Set Theory (ZF/ZFC)

Set Theory & Type Theory

Zermelo (1908), Fraenkel (1922), Skolem (1922) -- cumulative hierarchy

- V** All well-founded sets; cumulative hierarchy V_α for ordinal α
- G** Membership element-of; power set $P(x)$; union; separation; replacement; infinity axiom.
- 0** Ordinal level α ; patterns at level α cohere in V_α

PL INSIGHT

ZF builds mathematics from the empty set upward through a cumulative hierarchy indexed by ordinals: $V_0 = \text{empty}$, $V_{\alpha+1} = \text{power set of } V_\alpha$, $V_\lambda = \text{union of all } V_\alpha \text{ below } \lambda$. Russell's paradox is prevented because membership requires the member to exist at a strictly lower level than the containing set -- no set can be its own member. The axioms of Separation and Replacement constrain which collections can be formed: only those definable from already-existing sets. The Axiom of Choice (in ZFC) asserts that any collection of non-empty sets has a simultaneous choice function.

Forces

- + Foundation: no infinite membership descent
- + Extensionality: sets equal iff same members
- + With AC (ZFC): well-ordering theorem, Zorn's lemma

Breaks on

- ✗ No universal set: V is not a set (Russell's paradox prevented by level separation)
- ✗ Godel incompleteness: CH is undecidable in ZFC

Useful for

Foundations of mathematics Formal verification of mathematics (Metamath) Category theory foundations

30 New Foundations (NF / NFU)

Set Theory & Type Theory

W.V.O. Quine (1937) -- NF. Jensen (1969) -- NFU (consistent variant)

- V** All sets including a universal set $V = \{x : x=x\}$
- G** Stratified comprehension: $\{x : \phi(x)\}$ exists only if ϕ is stratified (typeable in type theory). Unrestricted otherwise.
- 0** 1.0 for consistent stratification levels

PL INSIGHT

NF avoids Russell's paradox not by stratifying the universe (as ZF does) but by adding a syntactic stratification condition to comprehension. A formula ϕ is stratified if you can assign a natural number to each variable so that membership always connects a type- n element to a type- $(n+1)$ set. Russell's set $\{x : x \text{ not-in } x\}$ fails stratification because x appears on both sides of membership at the same type level. The universal set $V = \{x : x=x\}$ passes because identity is trivially stratified. The restriction is on the formula describing the set, not on the rank of the set itself.

Forces

- + Universal set exists (unlike ZF)
- + Specker's theorem: AC is false in NF (AC disproves AC)
- + Strongly Cantorian sets behave classically

Breaks on

- ✗ NF consistency: open for full NF (NFU is consistent)
- ✗ AC fails in NF (Specker 1953)
- ✗ Lacks a well-ordering of all ordinals

Useful for

Alternative foundations study Cardinality without AC Theoretical exploration of foundational alternatives

31 Simple Theory of Types (STT / Church)

[Set Theory & Type Theory](#)

Russell (1908) ramified types; Church (1940) simple types

- V** Types: e (entities), t (truth values), $(a \rightarrow b)$ (functions). Values type-stratified.
- G** Lambda abstraction and application. Each gradient must respect type boundaries. Types prevent self-application.
- Θ** 1.0; load scales with type level

PL INSIGHT

Church's simple type theory assigns every expression a unique type from a hierarchy built from base types e (individuals) and t (truth values) and function types $a \rightarrow b$. Self-application ($\lambda x.x x$) cannot be typed: applying x to itself requires x to have both type a and type $a \rightarrow b$ simultaneously, which is impossible in any consistent type assignment. Russell's set $\{x : x \text{ not-in } x\}$ requires the membership predicate to have mutually incompatible types. Type-checking is decidable and every well-typed term reduces to a unique normal form.

Forces

- + Type safety: no self-application of untyped lambda
- + Russell's paradox prevented: R cannot be consistently typed
- + Decidability: type checking is decidable

Breaks on

- × No universal quantification over all types simultaneously
- × Higher-order types make equality complex

Useful for

Haskell/ML type theory foundations Higher-order logic (HOL) Interactive theorem provers Functional programming semantics

32 Martin-Lof Type Theory (MLTT)

[Set Theory & Type Theory](#)

Per Martin-Lof (1975, 1984) -- Intuitionistic type theory

- V** Dependent types: Π -types (dependent products), Σ -types (dependent sums). Types themselves are values.
- G** Constructive: all inhabitants are programs. Beta/eta reduction. Identity type $\text{Id}_A(a,b)$ inhabited iff $a = b$.
- Θ** Unbounded; each proof has finite load, different depths have different loads

PL INSIGHT

Dependent types allow the type of an expression to depend on the value of another. A Π -type $\Pi(x:A).B(x)$ is a function whose return type $B(x)$ varies with the argument x : to inhabit it you provide a function mapping each $a:A$ to a term of type $B(a)$. A Σ -type $\Sigma(x:A).B(x)$ is a dependent pair: an element $a:A$ together with a proof of $B(a)$. This makes types first-class mathematical objects -- propositions are types and proofs are their inhabitants. The universe hierarchy ($\text{Type}_0 : \text{Type}_1 : \text{Type}_2 : \dots$) prevents self-reference: each universe is itself a type, but of one level higher.

Forces

- + Propositions as types: proofs = programs (Curry-Howard)
- + Totality: all programs terminate
- + Π -types: proof of forall $x:A. B(x)$ is a function producing a Pattern of type $B(a)$ for every a

Breaks on

- × Impredicativity: $\text{Type}_0 : \text{Type}_0$ is inconsistent (Girard's paradox)
- × Uniqueness of identity proofs contradicted by HoTT

Useful for

Coq, Agda, Idris proof assistants Certified software Program extraction from proofs

33 Homotopy Type Theory (HoTT)

Set Theory & Type Theory

Voevodsky (2006-2010); Awodey & Warren (2009); Univalent Foundations Program (2013)

- V** Types as topological spaces. Paths between terms are homotopies. Higher-dimensional identity structure.
- G** Path induction (J-rule), transport, function extensionality, univalence axiom: $(A=B) = (A \simeq B)$ -- paths between types = equivalences.
- 0** Infinity-groupoid structure; load = homotopy dimension of path

PL INSIGHT

HoTT interprets types as topological spaces: terms are points, identity proofs are paths between points, proofs that two paths are equal are homotopies between paths, and so on through higher dimensions. The identity type $\text{Id}_A(a,b)$ may have multiple distinct inhabitants -- different proofs of equality that are not themselves provably equal. Voevodsky's Univalence Axiom states that type equivalence (an invertible function between types preserving all structure) is the same as type identity: isomorphic mathematical structures are literally equal, not merely analogous.

Forces

- + Univalence: isomorphic types are equal
- + Function extensionality (derivable from univalence)
- + Higher inductive types: circles, spheres definable as types

Breaks on

- ✗ Classical LEM undecidable (neither provable nor disprovable in HoTT)
- ✗ AC is undecidable (both consistent with HoTT)

Useful for

Formal foundations of mathematics Category theory Algebraic topology Cubical type theory (Lean 4, Agda-cubical)

34 Topos Logic (Elementary Topos / Lawvere)

Set Theory & Type Theory

F.W. Lawvere & Myles Tierney (1969-1972)

- V** Subobject classifier Ω . In **Set**: $\Omega = \{\text{false}, \text{true}\}$. In sheaves on X : $\Omega(U) = \{\text{sieves on } U\}$.
- G** $\text{True}: 1 \rightarrow \Omega$. AND, OR, NOT, IMP: $\Omega \times \Omega \rightarrow \Omega$. All defined by universal properties in the topos.
- 0** Context-dependent: each Grothendieck topology sets its own θ

PL INSIGHT

In an elementary topos, the subobject classifier Ω is a special object such that subobjects (generalised subsets) of any object A correspond bijectively to morphisms $A \rightarrow \Omega$. In the category **Set**, $\Omega = \{\text{false}, \text{true}\}$ and the internal logic is classical. In sheaves over a topological space X , $\Omega(U)$ consists of the open subsets of U -- truth at a local region U means truth on some open neighbourhood -- not every open set is either all of U or empty, so the logic is intuitionistic. Each topos determines its own logic through its Ω , and all connectives are defined by universal properties in the topos rather than by explicit truth tables.

Forces

- + Internal logic is always intuitionistic (LEM not forced in general)
- + Subobject classifier unique up to isomorphism
- + Mitchell-Benabou language: first-order logic internal to any topos

Breaks on

- ✗ LEM not forced in general toposes
- ✗ Classical logic only in Boolean toposes

Useful for

Algebraic geometry (etale cohomology) Continuous mathematics (smooth toposes)
Alternative foundations (Lawvere's synthetic differential geometry)

Georg Cantor (1883) Grundlagen einer allgemeinen Mannigfaltigkeitslehre; Gentzen (1936) consistency of PA

- V** Ordinals Ord: well-ordered sets up to order-isomorphism. $0, 1, 2, \dots, \omega, \omega+1, \dots, \omega \cdot 2, \dots, \omega^2, \dots, \omega^\omega, \dots, \varepsilon_0$ where $\varepsilon_0 = \sup\{\omega, \omega^\omega, \omega^{\omega^{\omega^{\dots}}}, \dots\}$.
- G** Addition: $\alpha + \beta$ = order type of α followed by β (not commutative: $1 + \omega = \omega \neq \omega + 1$). Multiplication: $\alpha \cdot \beta$ = α copies of β (not commutative). Exponentiation: ω^α . Cantor normal form: every ordinal $\alpha = \omega^{a_1} \cdot c_1 + \dots + \omega^{a_n} \cdot c_n$, $a_1 > \dots > a_n$, c_i finite.
- Θ** $\varepsilon_0 = \omega^{\omega^{\omega^{\dots}}}$: smallest fixed point of $\alpha \mapsto \omega^\alpha$. Proof-theoretic ordinal of Peano Arithmetic: transfinite induction up to ε_0 exactly captures PA's consistency strength.

PL INSIGHT

Ordinals measure the length of well-orderings. The Cantor normal form expresses every ordinal as a base- ω polynomial with decreasing exponents -- analogous to decimal notation but with infinite base. The ordinal ε_0 is the smallest fixed point of the exponentiation map $\alpha \mapsto \omega^\alpha$: it satisfies $\omega^{\varepsilon_0} = \varepsilon_0$ and cannot be reached from ω by finitely many exponentiations. Gentzen's theorem gives ε_0 a concrete logical meaning: transfinite induction up to ε_0 is exactly equivalent to the consistency of Peano Arithmetic. Below ε_0 , PA can verify well-foundedness; above it, PA cannot. This makes ε_0 the proof-theoretic ordinal of PA -- the precise location of PA on the ladder of logical strength.

Forces

- + Unique Cantor normal form for every ordinal (analogous to unique decimal representation)
- + Well-ordering: every non-empty class of ordinals has a least member -- transfinite induction valid up to any ordinal
- + Addition and multiplication are associative but NOT commutative: $1 + \omega = \omega$ but $\omega + 1 > \omega$
- + Gentzen's theorem: consistency of PA is equivalent to transfinite induction up to exactly ε_0

Breaks on

- ✗ $1 + \omega = \omega \neq 0 + 1 = \omega + 1$: addition is not commutative (prefix by finite ordinal disappears into ω)
- ✗ $\omega \cdot 2 = \omega + \omega \neq 2 \cdot \omega = \omega$: multiplication not commutative either
- ✗ No largest ordinal: every ordinal has a strict successor (the 'next' well-ordering)

Useful for

- Proof theory (ordinal analysis of formal systems) Set theory (ordinal arithmetic and cardinal arithmetic)
- Well-founded recursion (termination proofs for programs) Gentzen's consistency proof for arithmetic
- Computability theory (Kleene's ordinal notation)

Mathematical Carriers

19 carriers

35 Differential Calculus (Dual Numbers / LH)

Mathematical Carriers

Leibniz (1684), Newton (1687). Dual numbers: Clifford (1873). Autodiff: Wengert (1964).

- V** Dual numbers: $\{(r, \text{eps}) : r, \text{eps} \in \mathbb{R}, \text{eps}^2=0\}$. Extended real line.
- G** AND load rule = Leibniz product rule: $D(f*g) = Df*g + f*Dg$. Chain rule: $D(f(g)) = f'(g)*Dg$. All arithmetic via dual number extension.
- Θ** Machine epsilon ($\sim 1e-12$): derivative computed to within floating-point precision

PL INSIGHT

Dual numbers extend the reals with a nilpotent element eps satisfying $\text{eps}^2=0$. Computing $f(a+b*\text{eps})$ yields $f(a) + f'(a)*b*\text{eps}$ exactly -- the eps -component carries the derivative of f at a . The product rule is forced by the algebra: $(a+b*\text{eps})*(c+d*\text{eps}) = ac + (ad+bc)*\text{eps}$ since $\text{eps}^2=0$. Chain rule: $f(g(a+\text{eps})) = f(g(a)+g'(a)*\text{eps}) = f(g(a)) + f'(g(a))*g'(a)*\text{eps}$. These rules are arithmetic consequences of the dual number structure, not separate definitions. Every modern automatic differentiation framework (JAX, PyTorch, TensorFlow) implements this algebra.

Forces

- + Product rule: exactly forced by $\text{eps}^2=0$ boundary condition
- + Chain rule: forced by composition of AND operations
- + FTC: G_{diff} and G_{int} are inverse gradients

Breaks on

- × Non-smooth functions: $|x|$ at $x=0$ gives multiple subgradients
- × Stochastic processes: Ito calculus requires Stratonovich correction

Useful for

Scientific computing Neural network training (backpropagation) Physics simulation Optimization Sensitivity analysis

36 Complex Analysis (C-carrier)

Mathematical Carriers

Cardano (1545), Euler (1748), Cauchy (1821), Riemann (1851)

- V** $\mathbb{C} = \mathbb{R}^2$ with rotation gradient $G_{\text{rot}}^2 = G_{\text{neg}}$
- G** Complex multiplication: $(a,b)*(c,d) = (ac-bd, ad+bc)$. Cauchy-Riemann gradients. Contour integration. Holomorphic functions satisfy $d/dx = i*d/dy$.
- Θ** Analytic continuation: any finite radius of convergence

PL INSIGHT

The complex numbers embed rotation into arithmetic: multiplication by i rotates by 90 degrees, and $i^2=-1$ makes two quarter-turns a negation. Holomorphic functions satisfy the Cauchy-Riemann equations -- the function's rate of change is consistent under all rotation angles of the differentiation direction. This consistency forces a remarkable rigidity: complex differentiability at every point in a region implies infinite differentiability and analyticity. The Cauchy integral theorem follows: holomorphic functions have conservative gradient fields, so integrals around closed contours vanish.

Forces

- + Cauchy-Riemann equations: forced by complex differentiability
- + Cauchy integral theorem: gradient integral over closed contour = 0
- + Holomorphic implies analytic: complex differentiability implies infinitely differentiable

Breaks on

- × Non-holomorphic functions: $|z|^2$ differentiable as real but not as complex
- × Branch cuts: $\log(z)$ requires choice of branch

Useful for

Fluid dynamics Electromagnetism Quantum mechanics (amplitudes) Fourier analysis Riemann zeta function and number theory

Abraham Robinson (1966) -- Non-standard Analysis. Rigorous infinitesimals.

- V** $*R = R$ extended with infinitesimals (epsilon: $0 < \text{epsilon} < r$ for all positive r in R) and infinite numbers
- G** Standard part function: $\text{st}(*x)$ projects to nearest real. Transfer principle: any first-order statement true in R is true in $*R$.
- 0** Monad around 0: $\{x \text{ in } *R : |x| < r \text{ for all positive } r \text{ in } R\}$

PL INSIGHT

Robinson's hyperreal field $*R$ contains the standard reals R as a subfield, along with genuine infinitesimals eps (positive but smaller than every positive real) and infinite numbers H (larger than every real). The transfer principle guarantees that every first-order statement true in R remains true in $*R$: arithmetic, algebra, and basic analysis transfer completely. Calculus proceeds with actual infinitesimals (dy/dx as a literal ratio) rather than limit definitions. The standard part $\text{st}(x)$ maps each finite hyperreal to the unique real it differs from by an infinitesimal, making the limit operation an algebraic projection.

Forces

- + Transfer principle: first-order arithmetic transfers between R and $*R$
- + Saturation: any descending family of internal sets has non-empty intersection

Breaks on

- x External sets (like standard naturals N in $*N$) are not internal
- x Cannot refer to 'the set of infinitesimals' internally

Useful for

Rigorous infinitesimal calculus (Keisler calculus) Hyperfinite probability theory Ultraproduct constructions

Kurt Hensel (1897). Completion of $\mathbb{Q}$ under p -adic absolute value.

- V** $\mathbb{Q}_p$: completion of $\mathbb{Q}$ with respect to p -adic absolute value $|x|_p = p^{-v_p(x)}$
- G** p -adic valuation: $v_p(p^n \cdot a/b) = n$ when p does not divide a or b . Arithmetic via p -adic expansion. Distance: $d_p(x,y) = |x-y|_p$.
- 0** $1/p$: the unit ball in $\mathbb{Q}_p$

PL INSIGHT

The p -adic absolute value reverses ordinary intuitions of size: an integer is p -adically small when it is highly divisible by p . Specifically, $|p^n \cdot m|_p = p^{-n}$ when p does not divide m . This metric completion of $\mathbb{Q}$ gives $\mathbb{Q}_p$ an ultrametric: $d(x,z) \leq \max(d(x,y), d(y,z))$, stronger than the triangle inequality. Every triangle in $\mathbb{Q}_p$ is isosceles -- the two longest sides are equal. Arithmetic series converge in $\mathbb{Q}_p$ when terms shrink by divisibility, not by absolute magnitude: $1+p+p^2+\dots$ converges to $1/(1-p)$ in $\mathbb{Q}_p$ but diverges in $\mathbb{R}$.

Forces

- + Ultrametric inequality: $d_p(x,z) \leq \max(d_p(x,y), d_p(y,z))$ -- stronger than triangle inequality
- + Every triangle in $\mathbb{Q}_p$ is isosceles
- + Ostrowski's theorem: all completions of $\mathbb{Q}$ are either $\mathbb{R}$ or $\mathbb{Q}_p$

Breaks on

- x $\mathbb{Q}_p$ and $\mathbb{R}$ are incompatible: no isomorphism
- x p -adic integers are compact but have no standard order

Useful for

Number theory (local-global principle) Cryptography (lattice-based systems) Error-correcting codes

39 Tropical Arithmetic (Min-Plus Semiring)

Mathematical Carriers

Cuninghame-Green (1960s); named tropical by French mathematicians (1980s); Maslov (1987)

V $\mathbb{R} \cup \{+\infty\}$, with addition = min and multiplication = plus
G Tropical ADD $(a,b) = \min(a,b)$. Tropical MUL $(a,b) = a+b$. ADD-identity = $+\infty$. MUL-identity = 0.
0 0 (the MUL identity / additive seed)

PL INSIGHT

The tropical semiring replaces ordinary addition and multiplication with min and plus: $a (+) b = \min(a,b)$ and $a (*) b = a+b$. These satisfy the distributive law: $a + \min(b,c) = \min(a+b, a+c)$. Tropical matrix multiplication computes $(A*B)_{ij} = \min_k (A_{ik} + B_{kj})$, which is exactly Bellman's shortest-path equation -- tropical matrix powers give shortest paths of increasing step count. The semiring is idempotent ($\min(a,a)=a$) and has no subtraction. Tropical polynomials are piecewise-linear functions; tropical algebraic geometry studies their combinatorial corner loci which carry the essential structure of classical algebraic varieties.

Forces

- + Idempotency: $\min(a,a) = a$
- + Distributivity: $a + \min(b,c) = \min(a+b, a+c)$
- + Tropical matrix multiplication = shortest path computation

Breaks on

- × No subtraction: min has no inverse
- × No division (other than by MUL)

Useful for

Shortest paths (Dijkstra = tropical matrix multiplication) Scheduling and combinatorial optimization
Algebraic geometry (Newton polygons, amoebas) Phylogenetics

40 Quantum Mechanics (Density Matrix Carrier)

Mathematical Carriers

von Neumann (1932) -- *Mathematische Grundlagen*. Dirac (1930) formalism.

V Density matrices: positive semidefinite Hermitian ρ with $\text{Tr}(\rho)=1$. Pure states: $\rho = |\psi\rangle\langle\psi|$.
G Schrodinger evolution: $G_U(\rho) = U \rho U^\dagger$ (unitary, load-preserving). Measurement: $G_M(\rho) = M \rho M^\dagger / \text{Tr}(M \rho M^\dagger)$.
0 1.0 for pure states; purity $\text{Tr}(\rho^2)$ measures coherence

PL INSIGHT

A density matrix ρ is a positive semidefinite Hermitian operator with $\text{Tr}(\rho)=1$. Pure states satisfy $\text{Tr}(\rho^2)=1$ (rank-1 projectors $|\psi\rangle\langle\psi|$); mixed states have $\text{Tr}(\rho^2) < 1$ and arise from entanglement or incomplete knowledge. Unitary evolution $\rho \rightarrow U\rho U^\dagger$ is reversible and preserves all eigenvalues. Measurement by operator M produces an outcome with probability $\text{Tr}(M\rho M^\dagger)$ and updates the state to $M\rho M^\dagger / \text{Tr}(M\rho M^\dagger)$ -- irreversible. The no-cloning theorem follows from linearity: a cloning map cannot be linear unless the states being cloned are mutually orthogonal.

Forces

- + Born rule: probabilities = $\text{Tr}(M \rho M^\dagger)$
- + Unitarity: Schrodinger evolution is load-preserving (reversible)
- + Decoherence increases entropy (raises load)

Breaks on

- × Measurement collapses superposition: carrier changes from quantum to classical $\{0,1\}$
- × Non-commutative: observables A and B generally $AB \neq BA$ (Heisenberg)

Useful for

Quantum computing Quantum cryptography (BB84) Quantum error correction Quantum sensors

41 Information Theory (Shannon Carrier)

Mathematical Carriers

Claude Shannon (1948) -- A mathematical theory of communication

- V** $[0,1]$ for probabilities; R_+ for entropy. Probability distributions over finite/infinite alphabets.
- G** Entropy: $H(P) = -\sum p_i \log p_i$. Mutual information: $I(X;Y) = H(X) + H(Y) - H(X,Y)$. KL divergence: $D_{KL}(P||Q) = \sum p_i \log(p_i/q_i)$. Channel capacity: $C = \max_P I(X;Y)$.
- Θ** Channel capacity C bits/symbol

PL INSIGHT

Shannon entropy $H(P) = -\sum p_i \log_2 p_i$ quantifies the average uncertainty per outcome, equal to the minimum bits per symbol for lossless compression. The data processing inequality $I(X;Z) \leq I(X;Y)$ for any Markov chain $X \rightarrow Y \rightarrow Z$ formalizes that processing can only lose information -- no deterministic or stochastic transformation of Y can increase its mutual information with X . Channel capacity $C = \max_P I(X;Y)$ is the supremum over input distributions; the channel coding theorem guarantees reliable transmission at any rate $R < C$ and impossibility at $R > C$.

Forces

- + Source coding theorem: $H(X)$ bits needed per symbol to compress X
- + Channel coding theorem: reliable communication at rate $< C$ is possible
- + Data processing inequality: $I(X;Z) \leq I(X;Y)$ for Markov chain $X \rightarrow Y \rightarrow Z$

Breaks on

- × Zero-error information theory is much harder
- × Continuous channels require differential entropy (can be negative)

Useful for

- Data compression (gzip, Huffman, arithmetic coding)
- Error-correcting codes (Reed-Solomon, LDPC)
- Network capacity planning
- Cryptography foundations

42 Statistical Mechanics (Gibbs / Boltzmann Carrier)

Mathematical Carriers

Boltzmann (1872), Gibbs (1902). Entropy $S = k_B \ln(W)$.

- V** Microstates ω in phase space Γ . Probability distributions $p(\omega)$. Macrostates as equivalence classes.
- G** Liouville equation (Hamiltonian flow, load-preserving). Entropy gradient: $S = -k_B \sum p \ln p$. Gibbs measure: $p(\omega) = \exp(-E/k_{BT}) / Z$.
- Θ** $k_B T$ (thermal energy = natural load unit)

PL INSIGHT

Statistical mechanics connects microscopic dynamics to macroscopic thermodynamics via probability distributions over microstates. At thermal equilibrium the Boltzmann distribution $p(\omega) = \exp(-E(\omega)/k_{BT})/Z$ maximizes entropy subject to fixed average energy. The partition function $Z = \sum \exp(-E/k_{BT})$ encodes all thermodynamic properties: free energy $F = -k_{BT} \ln Z$, entropy $S = -dF/dT$, pressure $p = -dF/dV$. Boltzmann's H-theorem proves $dS/dt \geq 0$ for closed systems: the entropy-maximizing equilibrium distribution is an attractor of the dynamics.

Forces

- + Boltzmann H-theorem: $dS/dt \geq 0$ (entropy never decreases)
- + Equipartition: each quadratic degree of freedom gets $k_{BT}/2$ energy
- + Gibbs paradox: entropy of mixing is extensive

Breaks on

- × Time reversal: laws are symmetric but entropy is not (Boltzmann's paradox)
- × Fluctuations: entropy can locally decrease in small systems

Useful for

- Materials science
- Chemistry (reaction rates via Arrhenius)
- Molecular dynamics
- Thermodynamics of computation

43 Fisher Information (Statistical Manifold)

Mathematical Carriers

R.A. Fisher (1925). Rao (1945). Amari (1985) -- information geometry.

V Statistical manifold M : smooth family of probability distributions $\{p(x;\theta)\}$

G Fisher metric: $F_{ij}(\theta) = E[(\frac{d \log p}{d \theta_i})(\frac{d \log p}{d \theta_j})]$. Natural gradient: $F^{-1} \text{grad } L$.

Θ Fisher information F ; higher F = steeper gradient demand per unit parameter change

PL INSIGHT

Fisher information $F_{ij}(\theta) = E[\frac{d \log p(x;\theta)}{d \theta_i} \cdot \frac{d \log p(x;\theta)}{d \theta_j}]$ measures how sensitively the distribution $p(x;\theta)$ responds to parameter changes at θ . The Cramer-Rao bound $\text{Var}(\text{estimator}) \geq F(\theta)^{-1}$ sets the minimum achievable variance for any unbiased estimator. Geometrically, F defines the Riemannian metric on the manifold of probability distributions. The natural gradient $F^{-1} \cdot \text{grad}(L)$ moves parameters in the direction of steepest descent with respect to this intrinsic geometry, correcting the distortion that flat Euclidean gradient descent ignores.

Forces

+ Cramer-Rao bound: $\text{Var}(\text{estimator}) \geq F^{-1}$

+ Rao-Blackwell: sufficient statistic achieves Cramer-Rao bound

+ Natural gradient = Riemannian gradient with Fisher metric

Breaks on

× Singular Fisher information: degenerate models (non-invertible)

× Non-parametric: Fisher information is infinite

Useful for

Optimal statistical estimation

Neural network training (natural gradient methods)

Reinforcement learning (TRPO, PPO use Fisher metric)

55 Category Theory (Functors & Natural Transformations)

Mathematical Carriers

Eilenberg & Mac Lane (1945) General Theory of Natural Equivalences

V Objects $\text{Ob}(C)$ and morphisms $\text{Mor}(C)$ with composition $g \circ f: A \rightarrow C$ whenever $f: A \rightarrow B$, $g: B \rightarrow C$. Identity morphisms id_A for every object A .

G Functor $F: C \rightarrow D$ preserving composition ($F(g \circ f) = F(g) \circ F(f)$) and identities ($F(\text{id}_A) = \text{id}_{F(A)}$). Natural transformation $\eta: F \rightarrow G$ assigning each object A a morphism $\eta_A: F(A) \rightarrow G(A)$ such that naturality squares commute.

Θ Functoriality: $F(\text{id}_A) = \text{id}_{F(A)}$ and $F(g \circ f) = F(g) \circ F(f)$. An adjunction $F \dashv G$: bijection $\text{Hom}_D(F(A), B) \cong \text{Hom}_C(A, G(B))$ natural in A and B .

PL INSIGHT

A category defines objects entirely by their morphisms: internal structure is irrelevant, only relationships matter. The Yoneda lemma makes this precise -- every object A is completely determined by the presheaf $\text{Hom}(-, A)$, the functor assigning to each X the set of morphisms $X \rightarrow A$. Two objects with naturally isomorphic hom-functors are themselves isomorphic. Functors are structure-preserving maps between categories; natural transformations are maps between functors. An adjunction $F \dashv G$ establishes a canonical bijection between two sets of morphisms, unifying dozens of mathematical constructions (free-forgetful, product-diagonal, limit-constant) under one algebraic pattern.

Forces

+ Yoneda lemma: every object A is completely determined by $\text{Hom}(-, A)$; a natural isomorphism $\text{Hom}(-, A) \cong \text{Hom}(-, B)$ forces $A \cong B$

+ Composition is associative and identity morphisms act as two-sided units

+ Every adjunction $F \dashv G$ gives a monad $T = G \circ F$ on C and a comonad on D

+ Limits and colimits characterize universal constructions (products, coproducts, equalizers) via representable functors

Breaks on

× Not every functor has an adjoint (adjoint functor theorem requires solution-set condition)

× Not every category is locally small; size issues arise with the category of all sets

× Categorical equivalence does not imply isomorphism: two categories can be equivalent without being isomorphic

Useful for

Universal algebra (limits, colimits, Kan extensions)

Haskell and Scala type systems (functors, monads)

Homological algebra (derived functors, Ext, Tor)

Algebraic geometry (sheaves, topoi)

Type theory (categorical semantics, Curry-Howard-Lambek)

Évariste Galois (1832, posthumous 1846); Abel (1824); formalized by Dedekind & Weber (1882)

- V** Splitting field K of polynomial f over base field F ; Galois group $\text{Gal}(K/F)$ = field automorphisms of K fixing F pointwise. Intermediate fields $F \subseteq E \subseteq K$.
- G** Galois correspondence: subgroups $H \leq \text{Gal}(K/F) \leftrightarrow$ intermediate fields $E = K^H$ (fixed field of H). $H \rightarrow K^H$ is order-reversing: larger subgroup fixes fewer elements. $E \rightarrow \text{Gal}(K/E)$ is the inverse map.
- Θ** Degree $[K:F] = |\text{Gal}(K/F)|$ for Galois extensions. Solvability threshold: $\text{Gal}(K/F)$ must be a solvable group (subnormal series with cyclic quotients) for f to be solvable by radicals.

PL INSIGHT

Galois theory converts the question 'can this polynomial be solved by radicals?' into pure group theory. The Galois group $\text{Gal}(K/F)$ acts on the roots of f , and its structure determines which combinations of field operations reach every root from the base field. Solvability by radicals corresponds exactly to the Galois group being solvable -- having a subnormal series with cyclic quotients, meaning the symmetry can be peeled away in abelian layers. The Galois correspondence is order-reversing: a larger subgroup (more symmetry) fixes fewer field elements (smaller fixed field). The theorem that degree ≥ 5 polynomials are generically unsolvable by radicals follows because A_5 , the Galois group of a generic quintic, is simple and therefore not solvable.

Forces

- + Fundamental theorem: subgroup lattice of $\text{Gal}(K/F)$ is anti-isomorphic to intermediate field lattice
- + $|\text{Gal}(K/F)| = [K:F]$ for Galois (normal, separable) extensions
- + Normal subgroup $H \trianglelefteq \text{Gal}(K/F) \leftrightarrow$ the subextension K^H/F is itself Galois
- + Solvability by radicals $\leftrightarrow \text{Gal}(K/F)$ is a solvable group (Galois's theorem)

Breaks on

- ✗ Degree ≥ 5 polynomials with Galois group A_n or S_n ($n \geq 5$) are not solvable by radicals: A_5 is simple
- ✗ Inseparable extensions in characteristic $p > 0$ break the classical bijection
- ✗ Infinite Galois theory requires profinite topology on the Galois group

Useful for

- Impossibility proofs (trisecting angles, squaring the circle, quintic formula)
- Algebraic number theory (class field theory)
- Cryptography (elliptic curves over F_p)
- Coding theory (cyclic codes, BCH codes)
- Geometric construction theory

Henri Lebesgue (1902) *Intégrale, longueur, aire*; Borel (1898) *sigma-algebras*

- V** Sigma-algebra of measurable subsets of $\mathbb{R}^n$ with measure $\mu: \mathbb{A}[0, +\infty]$ satisfying sigma-additivity. L^p spaces: equivalence classes of measurable functions with finite p -th moment.
- G** Lebesgue integral $\int f \, d\mu$: supremum of integrals of simple functions approximating f from below. Radon-Nikodym theorem: if $\mu \ll \nu$ (absolute continuity) then $d\mu/d\nu$ is a measurable density function. L^p norm: $\|f\|_p = (\int |f|^p \, d\mu)^{1/p}$.
- Θ** Sigma-finiteness: $\mathbb{R}^n$ decomposes into countably many finite-measure sets (required for Radon-Nikodym). Completeness of L^p : Cauchy sequences converge in L^p .

PL INSIGHT

Lebesgue measure partitions the range (function values) rather than the domain. To integrate f , partition the y -axis into small intervals $[y, y+dy]$, measure the set $\{x: f(x) \approx y\}$ using μ , and sum $y \cdot \mu(\{x: f(x) \approx y\})$. This reversal allows integrating highly discontinuous functions (characteristic function of the rationals has Lebesgue integral 0 but no Riemann integral) and makes limit-integral interchange provable under minimal conditions. The Radon-Nikodym theorem shows that when one measure is absolutely continuous with respect to another ($\mu \ll \nu$), their relationship is encoded by a single measurable function $d\mu/d\nu$ -- the density, which is itself an element of L^1 .

Forces

- + Sigma-additivity: $\mu(\cup A_n) = \sum \mu(A_n)$ for pairwise disjoint measurable A_n
- + Monotone convergence: $f_n \nearrow f$ implies $\int f_n \, d\mu \rightarrow \int f \, d\mu$
- + Dominated convergence: $|f_n| \leq g$ with $\int g < \infty$ implies $\int f_n \rightarrow \int f$
- + Riesz-Fischer: L^p is complete (Cauchy sequences converge to an L^p function)

Breaks on

- ✗ Non-measurable sets exist assuming AC (Vitali sets, Banach-Tarski decomposition)
- ✗ The Lebesgue integral does not handle conditionally convergent improper integrals that Riemann does ($\int \sin(x)/x$ over $[0, \infty)$)
- ✗ Uncountable disjoint collections: sigma-additivity does not extend to uncountable families

Useful for

Probability theory (measure-theoretic foundations)

Fourier analysis (L^2 theory, Plancherel)

Functional analysis (L^p spaces, Hilbert spaces)

Partial differential equations (weak solutions)

Mathematical finance (stochastic integration)

Kiyosi Ito (1944) *stochastic integral*; Ito's lemma (1951); Stratonovich (1966) *alternative convention*

- V** Filtered probability space $(\Omega, \mathcal{F}_t, P)$; Brownian motion W_t ($W_0=0$, independent increments, $W_t - W_s \sim N(0, t-s)$); adapted processes H_t (measurable with respect to $\mathcal{F}_t$).
- G** Ito integral $\int H_s dW_s$: L^2 limit of left-Riemann sums (non-anticipating). Ito's lemma: for smooth f , $df(W_t, t) = (df/dt + \frac{1}{2} d^2f/dW^2)dt + (df/dW)dW_t$. SDE: $dX = \mu(X, t)dt + \sigma(X, t)dW$.
- Q** Quadratic variation $[W, W]_t = t$: the cumulative sum of squared Brownian increments equals t exactly, not zero as it would for smooth paths. The Ito correction is forced by this.

PL INSIGHT

Brownian paths have nonzero quadratic variation $[W, W]_t = t$, meaning $(dW)^2 = dt$ to first order rather than $(dt)^2$ as for smooth curves. This forces a modification of the chain rule: computing $d(f(W_t))$ requires keeping the second-order term $f''(W_t)(dW)^2 = f''(W_t)dt$ because it does not vanish. Ito's lemma $df = f'dW + \frac{1}{2}f''dt$ is an exact equality in L^2 , not an approximation. The stochastic integral $\int H dW$ is defined via left-endpoint Riemann sums (non-anticipating) precisely to ensure the integrand does not 'know' the future increments of W . This is the Ito correction: it appears whenever smooth functions are composed with Brownian motion.

Forces

- + Ito isometry: $E[\int H dW]^2 = E[\int H^2 dt]$ (the L^2 norm of the stochastic integral equals the L^2 norm of the integrand)
- + Quadratic variation $[W, W]_t = t$ (not zero): $(dW)^2 = dt$ to first order
- + Ito's lemma correction term $\frac{1}{2} d^2f/dW^2 dt$ is forced by $[W, W]_t = t$ and cannot be omitted
- + Ito processes form a semimartingale algebra closed under smooth functions

Breaks on

- ✗ Chain rule fails: $d(f(W_t)) \neq f'(W_t)dW_t$ -- the Ito correction $\frac{1}{2}f''(W_t)dt$ must be added
- ✗ Stratonovich calculus restores the chain rule but uses midpoint convention (anticipating integrals in definition)
- ✗ Non-adapted integrands require the Skorokhod integral (Malliavin calculus)

Useful for

- Mathematical finance (Black-Scholes equation, interest rate models)
- Statistical physics (Langevin equation, Fokker-Planck)
- Signal processing (Kalman filter)
- Population genetics (Wright-Fisher diffusion)
- Stochastic control (Hamilton-Jacobi-Bellman)

Grassmann (1844) exterior algebra; Cartan (1899) differential forms; de Rham (1931) cohomology

- V** Differential k -forms on smooth manifold M : smooth alternating multilinear maps $(TM)^k \rightarrow \mathbb{R}$. Space $\Omega^k(M)$. Zero-forms = smooth functions. n -forms on n -manifold = integrands.
- G** Exterior derivative $d: \Omega^k \rightarrow \Omega^{k+1}$ ($d(f) = df$, $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$). Wedge product $\wedge: \Omega^k \times \Omega^l \rightarrow \Omega^{k+l}$ (antisymmetric). Integration $\int_M \omega$ for n -forms over oriented n -manifolds.
- 0** $d^2 = 0$: the fundamental nilpotency identity, exact cancellation of mixed second partials via antisymmetry of wedge product.

PL INSIGHT

Differential forms provide a coordinate-free calculus of integration where $d^2 = 0$ is an exact algebraic identity, not a condition to be imposed. Mixed partial derivatives are symmetric ($d^2 f / dx dy = d^2 f / dy dx$) while wedge products are antisymmetric ($dx \wedge dy = -dy \wedge dx$), so every second-order term in $d^2 \omega$ appears twice with opposite signs and vanishes exactly. Stokes' theorem $\int_M d\omega = \int_{\partial M} \omega$ is then a single identity unifying all classical integral theorems. De Rham cohomology $H^k = \text{closed/exact}$ forms measures the global topology of M : non-trivial classes correspond to 'holes' that prevent local exactness from extending globally. The circle has $H^1 = \mathbb{R}$ (one hole); the torus has $H^1 = \mathbb{R}^2$ (two independent loops).

Forces

- + $d^2 = 0$: $d(d\omega) = 0$ for all forms -- mixed partial derivatives commute but wedge product is antisymmetric, so terms cancel exactly
- + Stokes' theorem: $\int_M d\omega = \int_{\partial M} \omega$ (unifying Green's theorem, Gauss's theorem, classical Stokes')
- + Poincaré lemma: locally every closed form ($d\omega = 0$) is exact ($\omega = d\alpha$); closed implies locally exact
- + de Rham cohomology $H^k = \ker(d_k) / \text{im}(d_{k-1})$ is a topological invariant of M

Breaks on

- ✗ Global exactness fails when M has non-trivial topology: $H^k(M) \neq 0$ means closed k -forms exist that are not globally exact
- ✗ Non-orientable manifolds require twisted coefficients (densities) rather than forms
- ✗ Non-smooth manifolds require distributional extensions (currents)

Useful for

Electromagnetism (Maxwell's equations as $dF = 0$, $d * F = J$)

General relativity (Einstein's equations in form language)

Algebraic topology (de Rham theorem: de Rham cohomology = singular cohomology)

Symplectic geometry

Integration on manifolds

Sophus Lie (1870s symmetry of ODEs); Killing (1888) classification; Cartan (1894) root systems

- V** Smooth manifold G with group structure where multiplication $G \times G \rightarrow G$ and inversion $G \rightarrow G$ are smooth. Lie algebra $\mathfrak{g} = T_e G$: tangent space at identity with Lie bracket $[X, Y]$.
- G** Lie bracket $[X, Y]$: infinitesimal commutator of flows generated by X and Y . Exponential map $\exp: \mathfrak{g} \rightarrow G$ integrating infinitesimal generators to group elements. Adjoint representation $\text{Ad}: G \rightarrow \text{GL}(\mathfrak{g})$. Killing form $B(X, Y) = \text{Tr}(\text{ad}_X \circ \text{ad}_Y)$.
- 0** Killing form: non-degenerate iff $\mathfrak{g}$ is semisimple (Cartan's criterion). $\exp$ is a local diffeomorphism near $0 \in \mathfrak{g}$. Simple Lie algebras classified: $A_n, B_n, C_n, D_n, E_6, E_7, E_8, F_4, G_2$.

PL INSIGHT

A Lie group packages continuous symmetry into an algebraic structure with a differential structure. The Lie algebra $\mathfrak{g}$ captures infinitesimal symmetries at the identity: each element X generates a one-parameter subgroup $\exp(tX)$ in G . The Lie bracket $[X, Y]$ measures the failure of X and Y 's flows to commute: if their flows commute then $[X, Y] = 0$. The exponential map integrates infinitesimal generators to finite group elements. Cartan's classification of simple Lie algebras -- $A_n (= \mathfrak{sl}_{n+1})$, $B_n (= \mathfrak{so}_{2n+1})$, $C_n (= \mathfrak{sp}_{2n})$, $D_n (= \mathfrak{so}_{2n})$, plus five exceptional algebras -- classifies all possible continuous symmetry groups up to local isomorphism.

Forces

- + Lie bracket antisymmetry: $[X, Y] = -[Y, X]$
- + Jacobi identity: $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$
- + $\exp(tX)\exp(sX) = \exp((t+s)X)$: one-parameter subgroups are group homomorphisms from $\mathbb{R}$
- + Lie's third theorem: every finite-dimensional real Lie algebra is the Lie algebra of a unique simply-connected Lie group

Breaks on

- ✗ Global topology not determined by algebra: $\text{SO}(3)$ and $\text{SU}(2)$ share Lie algebra $\mathfrak{su}(2)$ but $\text{SO}(3) = \text{SU}(2)/\mathbb{Z}_2$
- ✗ Non-compact groups have no faithful finite-dimensional unitary representations in general
- ✗ Infinite-dimensional Lie algebras (Kac-Moody) require completion and careful analysis

Useful for

- Particle physics (gauge symmetry: $\text{U}(1), \text{SU}(2), \text{SU}(3)$ in the Standard Model)
- Robotics ($\text{SO}(3)$ for 3D orientations, $\text{SE}(3)$ for rigid body motion)
- Differential equations (symmetry reduction via Lie's method)
- Representation theory
- Quantum groups and quantum computing

Poincaré (1895) *Analysis Situs*; Eilenberg-Steenrod axioms (1945); Eilenberg & MacLane (1945)

- V** Topological space X ; singular n -simplices: continuous maps $\sigma: \Delta^n \rightarrow X$. Free abelian chain groups $C_n(X) = \mathbb{Z}[\sigma_i]$ generated by n -simplices. Chain complex: $\dots \rightarrow C_2 \rightarrow C_1 \rightarrow C_0 \rightarrow 0$.
- G** Boundary map $d_n: C_n \rightarrow C_{n-1}$: $d_n(\sigma) = \sum (-1)^i \sigma|_{\{i\text{-th face}\}}$ (alternating sum of $(n-1)$ -faces). Homology $H_n(X) = \ker(d_n) / \text{im}(d_{n+1})$. Induced maps: $f: X \rightarrow Y$ continuous induces $H_n(f): H_n(X) \rightarrow H_n(Y)$.
- Θ** Betti numbers $b_n = \text{rank}(H_n(X; \mathbb{Z}))$: count independent n -dimensional holes. Euler characteristic $\chi(X) = \sum (-1)^n b_n$.

PL INSIGHT

A cycle is an n -chain with no boundary ($d(c)=0$); a boundary is an n -chain that bounds an $(n+1)$ -chain ($c=d(e)$). Since $d^2=0$, every boundary is automatically a cycle. The homology $H_n = \text{cycles}/\text{boundaries}$ counts cycles that are NOT boundaries -- these correspond to genuine topological holes of dimension n that enclose a void with no interior in X . The circle S^1 has $H_1 = \mathbb{Z}$ (one 1-hole). The torus has $H_1 = \mathbb{Z}^2$ (two independent loops). The sphere S^2 has $H_2 = \mathbb{Z}$ (one 2-hole). These algebraic invariants are topological: unchanged under any continuous deformation, and they can detect holes in high-dimensional spaces that have no visual representation.

Forces

- + $d^2=0$: $d_{n-1} \circ d_n = 0$ (codimension-2 faces cancel in pairs with opposite orientations)
- + Functoriality: $f: X \rightarrow Y$ continuous induces $H_n(f): H_n(X) \rightarrow H_n(Y)$ compatible with composition
- + Homotopy invariance: homotopy equivalent spaces have isomorphic homology groups
- + Long exact sequence of a pair: $\dots \rightarrow H_n(A) \rightarrow H_n(X) \rightarrow H_n(X, A) \rightarrow H_{n-1}(A) \rightarrow \dots$ exact

Breaks on

- ✗ Homology does not distinguish all homotopy types: lens spaces $L(7,1)$ and $L(7,2)$ have identical homology but are not homotopy equivalent
- ✗ Higher homotopy groups $\pi_n(X)$ for $n \geq 2$ contain information not captured by homology
- ✗ Homology with $\mathbb{Z}$ coefficients misses torsion information that cohomology with various coefficients can detect

Useful for

Computational topology (persistent homology for data analysis and topological data analysis) Knot theory (knot groups, Alexander polynomial)
Condensed matter physics (topological insulators, topological order) Robotics (configuration space topology) Network analysis

Fenchel (1949) conjugate duality; Kuhn & Tucker (1951) KKT conditions; Rockafellar (1970) convex analysis

- V** Convex subsets $C \subseteq \mathbb{R}^n$; convex functions $f: C \rightarrow \mathbb{R}$ satisfying $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$. Lagrangian $L(x, \lambda) = f(x) + \sum \lambda_i g_i(x)$.
- G** Subdifferential $\partial f(x)$: set of subgradients g with $f(y) \geq f(x) + (g, y-x)$ for all y . Fenchel conjugate $f^*(y) = \sup_x ((y, x) - f(x))$. KKT conditions: $0 \in \partial_x L$, $\lambda_i \geq 0$, $\lambda_i g_i(x) = 0$.
- Θ** Strong duality gap = 0: Slater's condition (existence of a strictly feasible point x° with $g_i(x^\circ) < 0$) guarantees that primal and dual optimal values coincide.

PL INSIGHT

Convex functions have no local minima that are not global: the subgradient optimality condition $0 \in \partial f(x)$ simultaneously identifies all global minimizers. Duality converts minimization over primal variables to maximization over Lagrange multipliers λ , and when strong duality holds (Slater's condition) both problems have the same optimal value and the duality gap is zero. The KKT conditions -- $0 \in \partial_x L$ and complementary slackness $\lambda_i g_i(x) = 0$ -- are simultaneously necessary and sufficient for optimality. Fenchel's double conjugate theorem $f^{**} = f$ (for closed convex f) shows that the conjugate construction is an exact involution on convex functions, giving a perfect duality between function values and slopes.

Forces

- + KKT conditions necessary and sufficient for optimality in convex programs (given constraint qualification)
- + Fenchel double conjugate: $f^{**}(x) = f(x)$ for closed convex f -- bi-conjugation recovers the function exactly
- + Subgradient exists everywhere for convex functions: $\partial f(x)$ is non-empty at every x in the domain
- + Strong duality under Slater: $\inf f = \sup_{\lambda \geq 0} \inf_x L(x, \lambda)$ when Slater's condition holds

Breaks on

- ✗ Non-convex problems: KKT necessary but not sufficient; local minima may not be global
- ✗ Infinite-dimensional optimization requires Banach space duality and may lack strong duality without additional conditions
- ✗ Strict complementarity may fail at degenerate optima (multiple active constraints)

Useful for

Machine learning (SVM, LASSO, neural network training via gradient methods) Operations research (linear/quadratic/conic programming)
 Economics (general equilibrium, mechanism design) Signal processing (compressed sensing, sparse recovery)
 Control theory (model predictive control)

Lagrange (1808); Hamilton (1833); Poincaré; Arnold (1974) *Mathematical Methods of Classical Mechanics*

- V** Symplectic manifold (M,ω) : smooth $2n$ -manifold with non-degenerate closed 2-form ω ($d\omega=0$). Standard: $\mathbb{R}^{2n}$ with $\omega=\sum dp_i\wedge dq_i$ (position-momentum phase space).
- G** Hamiltonian vector field X_H : unique field satisfying $i_{X_H}\omega=dH$ (contraction). Hamilton's equations: $\dot{q}_i=\partial H/\partial p_i$, $\dot{p}_i=-\partial H/\partial q_i$. Poisson bracket: $\{f,g\}=\omega(X_f,X_g)=\sum(\partial f/\partial q_i \partial g/\partial p_i - \partial f/\partial p_i \partial g/\partial q_i)$.
- Θ** Liouville's theorem: Hamiltonian flow preserves the symplectic volume form $\omega^n/n!$ (phase space is incompressible). Energy $H(q,p)=\text{const}$ along trajectories.

PL INSIGHT

Symplectic geometry is the natural language for Hamiltonian mechanics. The symplectic form ω on phase space encodes the fundamental Poisson bracket $\{q_i,p_j\}=\delta_{ij}$. Closedness $d\omega=0$ is not an assumption: in flat phase space $\omega=dp\wedge dq$ and $d^2=0$ automatically. Liouville's theorem -- Hamiltonian flows preserve the symplectic volume -- means phase space is incompressible: a statistical ensemble evolves without compression or expansion, preserving all volume elements. Noether's theorem translates every continuous symmetry of H into a conserved quantity via the Poisson bracket: time-translation gives energy conservation, rotation gives angular momentum. Darboux's theorem shows that all symplectic manifolds are locally identical -- there are no local symplectic curvature invariants.

Forces

- + Liouville's theorem: Hamiltonian flow preserves the symplectic volume form -- phase space is incompressible
- + Noether's theorem: $\{H,f\}=0 \Leftrightarrow f$ is a conserved quantity (symmetry $\leftrightarrow$ conservation law via Poisson bracket)
- + Darboux's theorem: every symplectic manifold is locally isomorphic to $(\mathbb{R}^{2n}, dp\wedge dq)$ -- no local symplectic invariants
- + Poincaré recurrence: any volume-preserving flow on a compact domain returns arbitrarily close to initial conditions

Breaks on

- ✗ Dissipative systems cannot be Hamiltonian: energy decrease violates Liouville's theorem
- ✗ Odd-dimensional manifolds cannot carry a symplectic structure (non-degeneracy requires even dimension)
- ✗ Groenewold-van Hove no-go theorem: no quantization can map all classical Poisson brackets to commutators

Useful for

- Classical mechanics (action-angle variables, integrable systems)
- Quantum mechanics (canonical quantization, deformation quantization)
- String theory (worldsheet action, symplectic reduction)
- Optical systems (ray optics as Hamiltonian flow)
- Toric geometry and mirror symmetry

John H. Conway (1976) *On Numbers and Games*; Knuth (1974) *Surreal Numbers* (popularization)

- V** Surreal numbers $No = \{L|R\}$ where L, R are sets of surreals with no $l \in L \geq 1 \ r \in R$. Includes all reals, all ordinals, infinitesimals, and transfinitely large/small numbers.
- G** Addition: $\{L|R\} + \{L'|R'\} = \{L + \{L'|R'\} \cup \{L|R\} + L' \mid R + \{L'|R'\} \cup \{L|R\} + R'\}$. Negation: $-\{L|R\} = \{-R \mid -L\}$. Multiplication: analogous using four extreme products. Simplicity rule: the canonical surreal between L and R is the one with earliest birthday (least ordinal rank).
- 0** Birthday of a surreal: the ordinal day on which it is born. 0 born day 0 ; ± 1 born day 1 ; all integers and rationals born by day ω ; infinitesimals born day $\omega+1$. Simpler = earlier = canonical.

PL INSIGHT

Surreal numbers are constructed by a universal inductive process starting from $\{\} = 0$ (nothing on either side). Day 1 produces $-1 = \{ \mid 0 \}$ and $1 = \{ 0 \mid \}$. By day ω , all integers and rationals have been born via the simplicity rule: the canonical surreal strictly between any two existing surreals is the simplest one. On day $\omega+1$, infinitesimals like $\{ 0 \mid 1, 1/2, 1/4, \dots \} = \epsilon$ are born -- positive but smaller than any positive rational. The simplicity rule makes the construction canonical: given any pair of sets L, R with no overlap, there is a unique simplest surreal between them. The resulting class No is the largest possible ordered field: every ordered field embeds into it.

Forces

- + No is a totally ordered field: addition, multiplication, and order all consistent with field axioms
- + Every ordinal is a surreal number: $\alpha = \{ \beta < \alpha \mid \}$ in the surreal construction
- + Real numbers embed isomorphically: the reals born at day ω are the standard real numbers
- + Simplicity rule: the canonical surreal between L and R is uniquely determined as the simplest (earliest-birthday) one

Breaks on

- ✗ No is a proper class, not a set -- standard ZFC set theory cannot handle it as a single object
- ✗ Standard analysis (limits, completeness) requires reformulation for transfinite surreals
- ✗ Cofinality issues arise for arithmetic operations on large surreals (infinite ordinal games)

Useful for

Combinatorial game theory (temperature theory in Go endgames; every game is a surreal number)

Non-standard analysis (alternative to Robinson's hyperreals)

Foundations of mathematics (universal ordered field)

Mathematical recreation (Conway's game theory)

44 Kleene Algebra (Regular Language Carrier)

Computer Science

Stephen Kleene (1956); axiomatised by Kozen (1994)

- V** Regular languages over alphabet Σ . Also: sets, relations, tropical semirings.
- G** PLUS (+) = union. TIMES (*) = concatenation. STAR (a*) = Kleene star = union of all n-fold concatenations.
- Θ** Finite automaton size (number of states)

PL INSIGHT

A Kleene algebra is an idempotent semiring ($a+a=a$) augmented with Kleene star satisfying $1+a^*a^* \leq a^*$ and the least fixed-point property: $ax \leq x$ implies $a^*x \leq x$. PLUS is union, TIMES is concatenation, STAR is zero-or-more-repetitions closure. Idempotency means taking the union of a language with itself adds nothing. Completeness: two regular expressions denote the same language if and only if they are provably equal in every Kleene algebra. Matrix Kleene algebra computes least fixed points of linear map equations, equivalent to solving reachability in finite automata.

Forces

- + Idempotency: $a+a = a$
- + STAR fixpoint: $a^* = 1 + a^*a^* = 1 + a^*a$
- + Completeness: Kleene algebra axioms are complete for regular languages

Breaks on

- ✗ Context-free languages: not captured by Kleene algebra
- ✗ Non-regular properties: pumping lemma limits expressibility

Useful for

Regular expression matching Model checking (bisimulation) Compiler theory Network protocol verification

45 Hoare Logic (Pre/Postcondition Carrier)

Computer Science

Tony Hoare (1969) -- An axiomatic basis for computer programming, CACM 12(10)

- V** Predicates over program states. $\{P\} C \{Q\}$: executing C in state satisfying P produces state satisfying Q .
- G** Hoare triples as gradients. Sequence: $\{P\} C1 \{R\}, \{R\} C2 \{Q\} \Rightarrow \{P\} C1;C2 \{Q\}$. Assignment: $\{P[e/x]\} x:=e \{P\}$.
- Θ** Partial vs total correctness: partial allows non-termination ($\theta=\infty$), total requires termination

PL INSIGHT

A Hoare triple $\{P\} C \{Q\}$ asserts partial correctness: starting from any state satisfying P and executing C to termination yields a state satisfying Q . Sequential composition threads via an intermediate predicate: $\{P\}C1\{R\}$ and $\{R\}C2\{Q\}$ give $\{P\}C1;C2\{Q\}$. The assignment axiom $\{P[e/x]\} x:=e \{P\}$ reasons backwards: substituting e for x in the postcondition gives the weakest precondition. The weakest precondition $wp(C,Q)$ is the weakest P making $\{P\}C\{Q\}$ valid -- mechanically computable for structured programs. Loop verification requires an invariant I : if I and cond together imply $wp(\text{body}, I)$, then $\{I\} \text{while cond do body} \{I \text{ and not-cond}\}$.

Forces

- + Weakest precondition: $wp(C,Q) = \text{weakest } P \text{ such that } \{P\}C\{Q\}$
- + Rule of consequence
- + Frame rule (separation logic extension)

Breaks on

- ✗ Unbounded loops: no total correctness proof
- ✗ Non-determinism: classical Hoare logic doesn't handle it well

Useful for

Program verification (Why3, VeriFast) Safety-critical software Compiler correctness proofs OS kernel verification

46 Separation Logic (Heap Carrier)

Computer Science

Reynolds (2002), O'Hearn & Pym (1999)

- V** Heaps: partial functions from addresses to values. Separating conjunction $P * Q$: P and Q hold for disjoint heap regions.
- G** Separating conjunction $*$: $P * Q = P$ holds on heap H_1 , Q on disjoint H_2 . Frame rule: $\{P\} C \{Q\} \Rightarrow \{P*R\} C \{Q*R\}$ for any R describing unchanged heap.
- Θ** Heap size; separation ensures non-interference between regions

PL INSIGHT

Separation logic extends Hoare logic with the separating conjunction $P*Q$: both P and Q hold simultaneously over non-overlapping, disjoint heap regions. Classical conjunction $P \text{ AND } Q$ allows P and Q to share heap cells; the separating conjunction requires complete disjointness. The frame rule $\{P\}C\{Q\}$ implies $\{P*R\}C\{Q*R\}$: a command accessing only the P -region cannot affect the R -region. The points-to assertion $l \mapsto v$ states that l is the only allocated location and holds value v . Memory safety -- no dangling pointer reads, no double frees -- becomes a syntactic validity check on separating conjunction.

Forces

- + Frame rule: local reasoning about heap regions
- + Separating conjunction is commutative and associative
- + Emp: empty heap is unit for $*$

Breaks on

- × Concurrent separation logic adds complexity
- × Permissions tracking requires additional structure

Useful for

Memory safety verification Concurrent program verification Rust's ownership system (formal semantics) OS memory management proofs

47 Lambda Calculus (Untyped / Simply Typed)

Computer Science

Alonzo Church (1932-1936); Church & Rosser (1936) confluence theorem

- V** Lambda terms (untyped); or typed terms with simple types (STLC)
- G** Beta reduction: $(\lambda x.M) N \rightarrow M[N/x]$. Alpha equivalence. Church encodings for data types.
- Θ** Normal form (termination): untyped LC may diverge ($\theta = \infty$); STLC always terminates

PL INSIGHT

Lambda calculus has two primitives: lambda abstraction ($\lambda x.M$ defines a function returning M with x free) and application ($M N$ applies M to N). Beta reduction $(\lambda x.M) N \rightarrow M[N/x]$ substitutes N for every free occurrence of x in M . The Church-Rosser theorem guarantees confluence: if a term reduces to two different terms, both reduce to a common term. The Y combinator $Y = \lambda f.(\lambda x.f(xx))(\lambda x.f(xx))$ satisfies $Y(f) = f(Y(f))$: it finds fixed points, enabling general recursion. Simply typed lambda calculus adds type annotations preventing self-application and guaranteeing all reductions terminate.

Forces

- + Church-Rosser (confluence): beta reduction is confluent
- + Curry-Howard: types = propositions, terms = proofs
- + Turing completeness of untyped LC

Breaks on

- × Self-application: $(\lambda x.xx)(\lambda x.xx)$ diverges (Y combinator)
- × Untyped: undecidable whether a term has a normal form

Useful for

Functional programming foundations (Haskell, ML) Type theory (Curry-Howard correspondence) Program semantics Proof assistants

48 Domain Theory (Scott Topology / CPO)

Computer Science

Dana Scott (1969, 1970) -- *mathematical semantics for lambda calculus*

- V** Complete partial orders (CPOs); Scott topology; continuous functions between CPOs. Bottom element bottom = undefined.
- G** Scott-continuous functions (preserve directed suprema). Fixed-point combinator: $Y(f) = \sup\{f^n(\text{bottom})\}$. Least fixed point.
- Θ** Computational approximation depth; bottom is the seed state with $L=0$

PL INSIGHT

A complete partial order (CPO) is a poset where every directed subset has a least upper bound. The bottom element represents 'undefined' -- the starting state of any computation. Scott-continuous functions preserve directed suprema: if increasing approximations converge to a limit, the function's values converge to the function at the limit. The least fixed point of $f:D \rightarrow D$ is $Y(f) = \text{join of } \{f^n(\text{bottom})\}$: starting with no information and iterating f builds up approximations. For recursive programs, $f^n(\text{bottom})$ is the result after n unfoldings of the recursion, and $Y(f)$ is the fully computed meaning.

Forces

- + Fixed-point theorem: every Scott-continuous $f:D \rightarrow D$ has a least fixed point
- + Approximation: every computable function is Scott-continuous
- + omega-chain completeness

Breaks on

- × Parallel-or is not Scott-sequential (requires continuity from both arguments)
- × Non-monotone functions are not representable

Useful for

Denotational semantics (Haskell's lazy evaluation) Recursive type theory Program analysis Fixed-point semantics of recursive programs

49 Abstract Interpretation (Galois Connection)

Computer Science

Cousot & Cousot (1977) -- *POPL. Systematic design of program analysis frameworks.*

- V** Abstract domain A with ordering $\leq_A$; concrete domain C with $\leq_C$. Galois connection: $\alpha(c) \leq_A a$ iff $c \leq_C \gamma(a)$.
- G** Abstraction $\alpha: C \rightarrow A$ (precision loss, load reduction). Concretization $\gamma: A \rightarrow C$. Abstract operations: widening for loop convergence.
- Θ** Precision of abstraction; widening threshold for termination

PL INSIGHT

Abstract interpretation pairs a concrete domain C (actual program states) with an abstract domain A (tractable approximations) via a Galois connection: $\alpha(c) \leq_A a$ iff $c \leq_C \gamma(a)$. The abstraction α maps concrete states to abstract ones (losing precision); concretization γ gives the most concrete meaning of an abstract state. Abstract operations must soundly overapproximate: for any concrete operation f , $\alpha(f(S)) \leq_A f_A(\alpha(S))$. This guarantees that any behaviour the abstract analysis misses was genuinely absent in the concrete program. False positives -- abstract states with no concrete counterpart -- are the price of decidable analysis.

Forces

- + Soundness: $\alpha(f(S)) \leq_A f_A(\alpha(S))$ -- abstract ops overapproximate concrete
- + Galois connection: α and γ are adjoint
- + Termination: widening ensures convergence

Breaks on

- × Precision loss: abstraction creates false positives
- × Precision vs termination: widening trades one for the other

Useful for

Static analysis (compiler warnings, buffer overflow detection) Model checking abstraction Security analysis (information flow) Type inference

Milner (1980) CCS; Hoare (1985) CSP; Bergstra & Klop (1984) ACP

- V** Processes P, Q as formal terms. Synchronization on channels. Bisimulation equivalence classes.
- G** Prefix: $\alpha.P$ (action α then P). Choice: $P+Q$ (load = min). Parallel: $P|Q$ (load = $L_P + L_Q$ where $C =$ synchronization cost). Restriction $P \backslash a$.
- 0** Bisimulation depth; number of steps to observation

PL INSIGHT

Process algebras define concurrent systems by structural induction: $\alpha.P$ (perform action α then become P), $P+Q$ (nondeterministic choice), $P|Q$ (parallel composition with synchronization on shared channels). Two processes P and Q are bisimilar ($P \sim Q$) when every action P can take to P' is matchable by Q to some Q' with $P' \sim Q'$, and vice versa. Bisimulation is stronger than trace equivalence: it captures the ability to respond to choices at each step, not just the set of possible sequences. Deadlock occurs when $P|Q$ reaches a state where both components are stuck waiting on each other.

Forces

- + Bisimulation: $P \sim Q$ if indistinguishable by any observer
- + Milner's expansion law
- + CCS axioms for guarded recursion

Breaks on

- ✗ Infinite processes: Turing-equivalent, undecidable in general
- ✗ Fairness requires machinery beyond bisimulation

Useful for

Concurrent and distributed systems Protocol verification Session types Modelling Go channels

Turing (1936) On Computable Numbers; Church (1936) lambda calculus; Kleene (1936) recursive functions

- V** Partial functions $N \rightarrow N$ (possibly undefined on some inputs). Turing machine descriptions encoded as natural numbers. Halting set $H = \{ \langle e, x \rangle : \text{TM}_e \text{ halts on input } x \}$.
- G** Universal TM U : given code e and input x , simulates TM_e on x . Reduction: $A \leq_m B$ (many-one) if there exists computable f with $x \in A \Leftrightarrow f(x) \in B$. Turing reduction $A \leq_T B$: A decidable using B as an oracle.
- $\emptyset$** Totality: a function is total computable iff it terminates on all inputs. Decidability threshold: A is decidable iff both A and its complement are recognizable (semi-decidable).

PL INSIGHT

Turing's key insight was that algorithms can be represented as data: a TM description is a string that can be given to a universal machine U as input. The diagonal argument then shows H is undecidable: suppose TM_d decides H ; run TM_d on the code $\langle d, d \rangle$ -- if TM_d says 'halts', then by the definition of TM_d it should not halt, contradiction; if TM_d says 'does not halt', it halts immediately, contradiction. Rice's theorem extends this to all semantic properties: any non-trivial fact about what a program computes is undecidable, because deciding it would let you solve the halting problem. The Church-Turing thesis asserts that 'effective computation' equals 'partial recursive function' -- justified by the coincidence of Turing machines, lambda calculus, and recursive functions.

Forces

- + Halting problem undecidable: diagonal argument shows no TM decides H (Turing 1936)
- + Rice's theorem: no non-trivial semantic property of TMs is decidable (semi-deciding TM behavior would solve H)
- + s-m-n theorem: computable functions of multiple arguments can be curried computably (parametric functions)
- + Recursion theorem: every computable transformation of TM codes has a fixed point (self-referential TMs exist)

Breaks on

- ✗ Halting problem, Post correspondence problem, Hilbert's 10th (Matiyasevich 1970), word problem for groups
- ✗ Turing degrees form a dense partial order: between any two comparable degrees lie infinitely many others
- ✗ Arithmetic hierarchy: Σ_n and Π_n sets are genuinely distinct for every n (no collapse)

Useful for

- Foundation of all computer science (what is computable?) Proving undecidability of natural mathematical problems
- Complexity theory (reductions between problems) Formal language theory and automata
- Mathematical logic (Godel's incompleteness via computability)

Cook (1971) NP-completeness of SAT; Karp (1972) 21 NP-complete problems; Razborov (1987) circuit lower bounds

- V** Decision problems (languages $L \subseteq \{0,1\}^*$). Complexity classes: P (poly-time deterministic), NP (poly-time verifiable), PSPACE (poly-space), BPP (randomized poly-time), #P (counting problems).
- G** Poly-time reduction $\leq_p$: $L_1 \leq_p L_2$ if computable f maps L_1 instances to L_2 instances in poly time. Boolean circuit: DAG of AND/OR/NOT gates; size=gate count, depth=longest path. NP-hardness: every NP problem $\leq_p L$.
- Θ** Polynomial = efficient (Cobham's thesis). P vs NP: can efficient verification (NP) always be converted to efficient search (P)?

PL INSIGHT

Computational complexity asks what resources are fundamentally necessary to solve computational problems, independent of specific machine details (up to polynomial factors). SAT is NP-complete: every problem in NP poly-time reduces to it, so if SAT has a poly-time algorithm then $P=NP$ and all NP problems become efficiently solvable. P vs NP asks whether finding a solution is always as easy as verifying one. Most cryptography depends on this gap: factoring, discrete log, and learning parity with noise are easy to verify but presumed hard to solve. The natural proofs barrier shows that most attempts to prove $P \neq NP$ are blocked by a deep obstruction: lower bound techniques powerful enough to prove circuit lower bounds would imply efficient pseudorandom generators do not exist -- a near-contradiction.

Forces

- + Time hierarchy: $DTIME(n^k) \subsetneq DTIME(n^{k+1})$ for each k -- more time strictly enables more computation
- + $P \subseteq NP \subseteq PSPACE \subseteq EXP$ (by simulation); $P \neq EXP$ by diagonalization
- + Cook-Levin: SAT is NP-complete -- every NP problem poly-time reduces to satisfiability of Boolean formulas in CNF
- + Ladner's theorem: if $P \neq NP$ then there exist problems in NP neither in P nor NP-complete

Breaks on

- ✗ P vs NP has been open since 1971 and is a Clay Millennium Prize Problem
- ✗ Relativization barrier: $P^A = NP^A$ for some oracles A , and $P^B \neq NP^B$ for others -- diagonalization cannot resolve P vs NP
- ✗ Natural proofs barrier (Razborov-Rudich): a large class of circuit lower bound techniques provably cannot separate P from NP

Useful for

- Cryptography (hardness assumptions underlie RSA, AES, elliptic curve cryptography)
- Algorithm design (approximation algorithms when NP-hard)
- Database query optimization
- AI hardness results (PSPACE-hard planning, NP-hard inference)
- Formal verification (SAT solvers, model checking)

Hilbert (1900) formalism program; Gentzen (1935) sequent calculus & consistency of PA; Takeuti, Schütte, Feferman (1960s-1990s)

- V** Formal proofs as syntactic trees in sequent calculus. Sequent $\Gamma \vdash \Delta$: from hypotheses Γ derive one of conclusions Δ . Cut rule: from $\Gamma \vdash A, \Delta$ and $\Gamma', A \vdash \Delta'$ derive $\Gamma, \Gamma' \vdash \Delta, \Delta'$.
- G** Cut elimination (Gentzen's Hauptsatz): every proof with cuts can be transformed into a cut-free proof. Ordinal assignment $|P|$ to proof P : measures complexity. Proof-theoretic ordinal $|T| = \sup$ of ordinals of provably terminating recursions.
- 0** Proof-theoretic ordinals: $|PA| = \epsilon_0$, $|ATR_0| = \Gamma_0$ (Feferman-Schütte ordinal), $|Pi_1^1-CA_0| = \Psi(\Omega_\omega)$. These are the precise consistency strengths.

PL INSIGHT

Proof theory analyzes proofs as mathematical objects. Cut elimination removes the 'lemma' rule: if you can prove something using an intermediate formula (cut formula A), you can always find a proof not mentioning A . The result may be astronomically longer but has the subformula property: every formula in the proof is a subformula of the statement being proved, giving complete transparency about what is being used. Ordinal analysis assigns a 'proof-theoretic ordinal' to each formal system -- the supremum of ordinals it can prove are well-founded. This ordinal is the exact threshold of the system's transfinite induction strength, placing formal systems precisely on the ladder of logical strength.

Forces

- + Hauptsatz (cut-elimination): any provable sequent has a cut-free proof (though possibly much longer)
- + Subformula property: every formula in a cut-free proof is a subformula of the end-sequent -- proofs are fully transparent
- + Godel's second incompleteness: no consistent sufficiently strong T can prove its own consistency ($\text{Con}(T)$) within T
- + Gentzen's result: transfinite induction up to ϵ_0 proves $\text{Con}(PA)$; induction below ϵ_0 is provable in PA

Breaks on

- ✗ Cut-free proofs can be non-elementarily longer than proofs using cuts (tower of exponentials blowup)
- ✗ Godel's incompleteness: T cannot prove its own consistency -- any such proof uses assumptions beyond T
- ✗ The omega-rule (if T proves $\phi(n)$ for every n , conclude $\forall x \phi(x)$) makes PA complete but non-recursive

Useful for

- Consistency proofs (Gentzen's proof that PA is consistent using ϵ_0 -induction)
- Foundations of mathematics (measuring axiom strength)
- Automated theorem proving (proof complexity, cut-free proofs for completeness)
- Formal verification (proof objects in proof assistants)
- Reverse mathematics (calibrating theorem strength)

Tarski (1936) truth definition; Gödel (1930) completeness; Henkin (1949) constants; Morley (1965) categoricity

- V** First-order structures $M=(D,I)$: non-empty domain D with interpretation I of constant/function/relation symbols. Satisfaction: $M \models [s]$ for variable assignment s , defined inductively.
- G** Tarski's truth definition: $M \models \phi$ by structural induction (atomic, negation, conjunction, quantification). Elementary equivalence: $M \equiv N$ iff same first-order sentences true. Ultraproduct: $\prod M_i / U$ for ultrafilter U on index set.
- Θ** Cardinality $|D|$; Löwenheim-Skolem: first-order theories cannot fix cardinality of infinite models. Morley: categoricity at one uncountable cardinal $\Leftrightarrow$ categoricity at all uncountable cardinals.

PL INSIGHT

Model theory studies the relationship between formal languages and their interpretations. The completeness theorem equates syntactic provability with semantic validity: a formula is provable from axioms iff it is true in every model of those axioms. The compactness theorem, its most powerful corollary, shows that if a property requires infinitely many axioms, no finite subcollection can express it. Löwenheim-Skolem reveals that first-order logic is deeply permissive about cardinality: the same theory can have countable and uncountable models. Non-standard models of arithmetic contain 'infinite integers' and models of analysis contain infinitesimals -- these are not pathologies but unavoidable features of first-order expressibility.

Forces

- + Completeness theorem (Gödel 1930): $\vdash \phi \Leftrightarrow \models \phi$ -- provability coincides exactly with semantic validity
- + Compactness: a set Σ of sentences has a model iff every finite subset has a model
- + Löwenheim-Skolem downward: any theory with an infinite model has a countable model
- + Löwenheim-Skolem upward: any theory with an infinite model has models of all larger cardinalities

Breaks on

- ✗ Gödel incompleteness: PA cannot prove all true statements about $\mathbb{N}$ -- first-order arithmetic is incomplete
- ✗ Non-standard models: any theory with an infinite model has non-standard (non-intended) models of all infinite sizes
- ✗ First-order logic cannot characterize $\mathbb{N}$ or $\mathbb{R}$ up to isomorphism -- they each have non-standard models

Useful for

- Non-standard analysis (Robinson's hyperreals via compactness and ultraproducts)
- Algebra (axiomatizing algebraically closed fields, real closed fields)
- Database theory (query equivalence, Codd's relational model)
- Philosophy of mathematics (what axioms characterize mathematical structures?)
- Automated reasoning (SAT solvers as model finders)

Harvey Friedman (1975); Simpson (1999) *Subsystems of Second-Order Arithmetic*

- V** Second-order arithmetic L_2 : formulas over natural numbers n and sets $X \subseteq \mathbb{N}$. Five subsystems: RCA_0 (computable sets), WKL_0 (+weak König lemma), ACA_0 (+arithmetical comprehension), ATR_0 (+arithmetic transfinite recursion), $\Pi_1^1-CA_0$ (+ Π_1^1 -comprehension).
- G** Formal equivalence over RCA_0 : theorem T is equivalent to axiom system S if both $T \Rightarrow S$ and $S \Rightarrow T$ are provable in RCA_0 . Reversing a theorem: proving the axiom FROM the theorem, not just the theorem from the axiom.
- Θ** Proof-theoretic strength: $RCA_0 < WKL_0 < ACA_0 < ATR_0 < \Pi_1^1-CA_0$. Ordinals: ω^ω , ω^ω , ε_0 , Γ_0 , $\Psi(\Omega_\omega)$. Strength measures exactly how much set-existence each system requires.

PL INSIGHT

Reverse mathematics asks which set-existence axioms are necessary and sufficient to prove a given theorem. The surprising finding is that almost all of classical analysis and combinatorics is provably equivalent over a weak base theory (RCA_0) to exactly one of five subsystems. 'Reversing' a theorem means proving the axiom from the theorem, establishing that the theorem is not just a consequence of the axiom but equivalent to it over RCA_0 . This reveals the exact logical content of mathematical theorems: Heine-Borel is equivalent to the statement that every infinite binary tree has a branch (WKL_0). The theorem tells you precisely which set-existence principles it requires.

Forces

- + The Big Five phenomenon: almost all classical theorems are equivalent over RCA_0 to exactly one of the five
- + WKL_0 is equivalent over RCA_0 to: Heine-Borel compactness, Brouwer fixed-point, separating hyperplane theorem
- + ACA_0 is equivalent to: Bolzano-Weierstrass, Ramsey for pairs, comparability of well-orderings
- + ATR_0 is equivalent to: Ulm's theorem, Lusin's theorem on Borel sets, comparability of countable well-orderings

Breaks on

- ✗ Some theorems fall between the Big Five: Ramsey for 4-tuples exceeds ACA_0 but falls below ATR_0
- ✗ Not all mathematics is covered: Π_2 sentences (Goldbach, Riemann hypothesis) lie in different hierarchies
- ✗ Proof-theoretic ordinals of the Big Five differ: ATR_0 has ordinal Γ_0 while $\Pi_1^1-CA_0$ has $\Psi(\Omega_\omega)$

Useful for

- Foundations of mathematics (what axioms are truly needed?) Computational mathematics (theorems with computable witnesses, in RCA_0)
- Proof theory (calibrating strength via ordinal analysis) Formal verification (identifying minimal axiom sets)
- Philosophy of mathematics (mathematical necessity)

Mac Lane (1971) monad as monoid in endofunctors; Moggi (1991) monads for computational effects; Wadler (1992) Haskell monads

- V** Endofunctor $T: C \rightarrow C$ with natural transformations $\eta: Id \Rightarrow T$ (unit/return) and $\mu: T^2 \Rightarrow T$ (multiplication/join). In Haskell: type constructor M with `return :: a -> M a` and `(>>=) :: M a -> (a -> M b) -> M b`.
- G** Kleisli composition `(>=>)` for effectful functions: `(f>=>g)(x)=f(x)>=>g`. Kleisli category $Kl(T)$: same objects as C , morphisms $A \rightarrow B = \text{morphisms } A \rightarrow T(B)$ in C . Every adjunction $F \dashv G$ gives a monad $T = G \circ F$.
- Θ** Three monad laws from $\mu \circ T(\eta) = id = \mu \circ \eta_T$ and $\mu \circ \mu_T = \mu \circ T(\mu)$: left identity (`return a>=>f=f a`), right identity (`m>=>return=m`), associativity (`((m>=>f)>=>g=m>=>(x->f x>=>g))`).

PL INSIGHT

A monad is a monoid object in the category of endofunctors: it packages a computational effect -- state, exceptions, I/O, non-determinism, probability -- as a type constructor T with two operations. Return wraps a pure value into the effectful context; bind `(>=>)` sequences two effectful computations by passing the result of the first into the second. The monad laws ensure sequencing is associative and return is a neutral element. This uniform interface captures any effect while keeping pure code pure. The Kleisli category of a monad absorbs effects into the morphism type -- every effectful computation becomes a pure morphism in a derived category where the effect is part of the structure.

Forces

- + Three monad laws from the category-theoretic axioms for a monoid in the monoidal category $[C, C]$ of endofunctors
- + Kleisli composition is associative and return is its two-sided unit
- + Every adjunction $F \dashv G$ generates a monad $T = G \circ F$ and a comonad $G \circ F$ on the opposite side
- + The Kleisli category $Kl(T)$ packages effectful computation: morphisms are pure functions into the effectful context

Breaks on

- ✗ Not every endofunctor is a monad: no natural unit or multiplication may exist
- ✗ Composing two monads does not give a monad in general -- requires a distributive law $\lambda: T'T \Rightarrow TT'$
- ✗ The monad laws are genuine algebraic laws that code must satisfy; violating them causes bugs

Useful for

- Functional programming (Haskell IO, Maybe, List, State, Parser, Probability monads)
- Denotational semantics of programming languages
- Effect systems and type-safe side effects
- Database query languages (list monad = SQL-like comprehensions)
- Category-theoretic logic and proof theory

Carl Adam Petri (1962) *Kommunikation mit Automaten* (PhD thesis, Darmstadt)

- V** Bipartite graph $N=(P,T,F,W,M_0)$: places P (local states), transitions T (events), arcs $F \subseteq (P \times T) \cup (T \times P)$, weights $W: F \rightarrow \mathbb{N}$, initial marking $M_0: P \rightarrow \mathbb{N}$ (token counts per place).
- G** Transition t enabled at marking M iff $M(p) \geq W(p,t)$ for all input places p . Firing: $M \rightarrow M'$ where $M'(p) = M(p) - W(p,t) + W(t,p)$. Reachability: $M' \text{ reachable } (M \rightarrow^* M')$ via firing sequences.
- Θ** Conservation laws: T -invariants (y with $F^T y = 0$) give token-preserving circuits. Place-invariants (x with $x^T F = 0$) give linear combinations of token counts that never change.

PL INSIGHT

Petri nets model concurrency by distributing tokens (resources/data) across places (local states) with transitions (events) firing whenever their input places have sufficient tokens. Firing is local: multiple transitions can fire simultaneously if they access disjoint sets of places. Conflict arises when two transitions share input places with insufficient tokens -- at most one can fire. This visual, mathematically precise model captures concurrency, parallelism, synchronization, and resource sharing without any global clock. Place-invariants are conservation laws: a weighted sum of token counts that never changes regardless of firing sequence, analogous to conserved quantities in physics.

Forces

- + T-invariant conservation: firing sequences corresponding to T-invariants return the net to its original marking
- + Place-invariant: $x^T M = x^T M_0$ for all reachable M (total weighted token count is constant along place-invariants)
- + Deadlock: reachable marking where no transition is enabled -- detectable by structural analysis
- + Liveness analysis: whether every transition can eventually fire from any reachable marking

Breaks on

- ✗ Reachability is decidable (Mayr 1981) but EXPSPACE-hard; in practice requires heuristics
- ✗ General Petri nets are Turing-complete: adding inhibitor arcs or reset arcs makes them undecidable
- ✗ Boundedness (token counts stay bounded) is also EXPSPACE-hard

Useful for

Workflow management systems (business process modeling)

Distributed systems verification

Biological pathway modeling (metabolic networks, gene regulatory networks)

Hardware verification (asynchronous circuits)

Manufacturing scheduling (flexible manufacturing systems)

paudit.py (2026), Propagation Logic applied to code; solver boundary due to Young (1950) and Ostrowski (1954)

- V** Contract-equivalence classes of program variants. Two programs belong to the same class iff they honour the same contract -- they produce the same outputs on a finite probe set. A point in V is 'a program up to contract'.
- G** Contract-preserving transforms whose closure IS contract preservation: hoisting, vectorising, re-ordering, choice of relaxation gradient (Jacobi / Gauss-Seidel / SOR). A legal gradient maps a variant to another variant in the same class.
- Θ** The budget and the contract. A variant that leaves the contract class, or whose measured load exceeds the budget, cannot cohere and is refused. Designation = 'inside the contract class'.

PL INSIGHT

Optimising code is Propagation Logic. The value space is every version of a program that computes the same thing (the contract-equivalence class); the gradients are the transforms that move between those versions; the load is what it costs to run, composing by max-plus so the total is the critical path; and the coherence threshold is the contract. To optimise is to descend the load by applying gradients without leaving the contract class. The auditor runs a campaign -- proposing transforms, measuring each, and gating: accept only if the contract holds AND the load drops. A refusal is not a failure but theta doing its job, declining a cheaper variant that left the class. Turned on a linear solver, the same frame recovers the classical result that the relaxation parameter re-gears the gradient without moving the solution, with a hard coherence wall at $\omega = 2$. A map that re-draws itself when the terrain moves, and refuses every road that breaks a bridge.

Forces

- + Max-plus load: sequential work adds ($L_{seq} = \text{sum}$), parallel work takes the max ($L_{par} = \text{max}$); the total cost equals the single longest chain, the critical path
- + The moving critical path: reducing the current critical task shrinks its path until a different path becomes longest -- the bottleneck moves, so the map must be re-drawn
- + Coherence boundary at $\omega = 2$ for SPD systems (Ostrowski): below it SOR converges, above it the residual grows -- decoherence, not slowness
- + Young's optimal relaxation $\omega^* = 2/(1 + \sin(\pi h)) = 1.9078$ predicts the measured sweet spot of the iteration-count sweep
- + Re-gearing G changes cost but not the fixed point: every accepted solver agrees with a direct reference solution to about one part in a million

Breaks on

- ✗ Finite probe set is not a proof of semantic equivalence -- the contract is a stipulation carried on every accept, not a theorem (STIPULATED)
- ✗ Gated local speedups do not compose into global optimality: Amdahl bounds each step and the moving critical path re-ranks the rest
- ✗ The auditor is not claimed to find ALL bottlenecks -- the profile is a finite sample (UNPAID)

Useful for

- Compiler and JIT optimisation passes verified against a contract
- Performance auditing with tamper-evident, claim-tiered reports
- Numerical solver tuning (choosing the relaxation gradient) with a measured coherence boundary
- Parallel scheduling and critical-path analysis (max-plus / tropical algebra)
- Regression-safe refactoring where the probe set is the acceptance test

51 DRAS Scale Carrier (Renormalization Group)

Physics

Wilson (1971) renormalization group; Callan-Symanzik equation; DRAS formalism (Pugmire 2026)

- V** (coupling α , scale E , load L). Continuous: $\alpha: R_+ \rightarrow R$ (running coupling as function of scale).
- G** G_{scale} : $v_{\text{new}} = v / (1 + \beta * \Delta(\ln E))$. $L_{\text{new}} = L + |\beta * \Delta(\ln E)|$. Beta function: $\beta(\alpha) = d\alpha / d(\ln E)$.
- Ø** Landau pole: $L \rightarrow \text{infinity}$ as denominator $\rightarrow 0$; UV cutoff Λ

PL INSIGHT

The renormalization group describes how coupling constants change with the energy scale E at which a quantum field theory is probed. The beta function $\beta(\alpha) = d\alpha / d(\ln E)$ governs this running: $\beta < 0$ (asymptotic freedom, as in QCD) means the coupling decreases at higher energies and the theory becomes weakly interacting at short distances. $\beta > 0$ means the coupling grows, eventually reaching a Landau pole -- a finite energy where perturbation theory diverges. The Callan-Symanzik equation ensures that physical observables are scale-independent even as bare couplings run.

Forces

- + Running coupling: α changes with scale E
- + Asymptotic freedom (QCD: $\beta < 0$)
- + Callan-Symanzik equation: scale invariance of observables

Breaks on

- ✗ Landau pole: coupling diverges at finite scale (QED, scalar ϕ^4)
- ✗ Triviality: some theories become non-interacting in the UV limit

Useful for

- QCD (asymptotic freedom)
- Particle physics phenomenology
- Condensed matter critical phenomena
- Wilson renormalization in statistical mechanics

52 Thermodynamic Carrier (Entropy / Landauer)

Physics

Carnot (1824), Clausius (1865), Boltzmann (1877), Landauer (1961), Berut et al. (2012) experimental verification

- V (S, E, N, V) -- entropy, energy, particle number, volume. Or (p, q) in phase space.
- G Hamiltonian flow: conservative, load-preserving. Entropy gradient: S increases under irreversible processes. Landauer: erasing 1 bit costs $k_B T \ln(2)$ at temperature T.
- Θ $k_B T$ (thermal energy per degree of freedom); $k_B T \ln(2)$ per bit erased

PL INSIGHT
Landauer's principle establishes that irreversible information erasure has a minimum thermodynamic cost: erasing one bit dissipates at least $k_B T \ln(2)$ of heat. This follows from the second law: erasing a bit decreases the entropy of the storage medium by $k_B \ln(2)$, and the second law requires this decrease to be compensated by heat released to the environment. Reversible operations (copying, reading without reset) have zero thermodynamic cost in principle. Maxwell's demon appears to violate the second law by selecting low-entropy microstates, but the demon must store measurement results and eventually erase its memory to operate cyclically -- the erasure cost restores the entropy balance.

- Forces**
- + Second Law: $dS/dt \geq 0$ in closed systems
 - + Landauer limit: $k_B T \ln(2)$ minimum per bit erased
 - + Equipartition: $k_B T/2$ per quadratic degree of freedom
- Breaks on**
- ✗ Maxwell's demon: apparent Second Law violation until demon's memory erased (Landauer resolves it)
 - ✗ Fluctuations: entropy can locally decrease in small systems
- Useful for**
- Thermodynamics of computation
 - Maxwell's demon resolution
 - Physical limits of AI (energy per inference)
 - Black hole thermodynamics (Bekenstein-Hawking entropy)

53 General Relativity (Tensor / Manifold Carrier)

Physics

Einstein (1915) -- Die Feldgleichungen der Gravitation. Riemann (1854) differential geometry.

- V (M, g) -- Lorentzian manifold M with metric tensor g. Tensor fields over M.
- G Covariant derivative: parallel transport, preserves tensor type. Einstein equations: $G_{\mu\nu} = 8\pi T_{\mu\nu}$ (gradient relation between curvature and stress-energy).
- Θ Curvature scale $1/R^2$; singularity: $\theta \rightarrow 0$ (load $\rightarrow$ infinity)

PL INSIGHT
General relativity identifies gravity with spacetime curvature. The Einstein field equations $G_{\mu\nu} = 8\pi T_{\mu\nu}$ relate the Einstein tensor (encoding spacetime curvature through the Riemann tensor) to the stress-energy tensor (mass, energy, momentum, and pressure). Free-falling objects follow geodesics -- curves of extremal proper time -- which is why gravity feels like acceleration: geodesic deviation is measurable curvature. The Bianchi identity automatically implies energy-momentum conservation. Singularities are where geodesic completeness fails -- the theory predicts its own breakdown at black hole cores and the Big Bang.

- Forces**
- + Bianchi identity: $\nabla_\mu G^{\mu\nu} = 0$ (energy-momentum conservation)
 - + Geodesic equation (load-minimizing paths)
 - + Equivalence principle: locally flat spacetime
- Breaks on**
- ✗ Singularities: load diverges at black holes and Big Bang
 - ✗ Quantum gravity: GR and QM carriers are incompatible at Planck scale
- Useful for**
- GPS (relativistic corrections required)
 - Black hole physics
 - Gravitational wave detection (LIGO)
 - Cosmology and dark energy modeling

Dirac (1927) quantum electrodynamics; Feynman, Schwinger, Tomonaga (1940s) renormalization

- V** Fock space $F = \text{direct sum of } H^n(x)$ (tensor products of single-particle Hilbert spaces). Creation $a^\dagger$, annihilation a operators.
- G** Feynman path integral: amplitude = integral of $\exp(iS/\hbar)$ over field configurations. Wick's theorem. LSZ reduction formula. Renormalization group.
- Θ** UV cutoff Λ ; coherence = renormalizability (finite load at each perturbative order)

PL INSIGHT

Quantum field theory describes particles as discrete excitations of underlying fields. The Fock space $F = \text{direct sum over } n \text{ of } H^n(\text{tensor } n)$ accommodates states with any definite particle number: the vacuum, one-particle states, two-particle states, and so on. Creation operator $a^\dagger(k)$ adds a particle with momentum k ; annihilation $a(k)$ removes one; they satisfy $[a(k), a^\dagger(k')] = \delta(k-k')$. Feynman's path integral formulates amplitudes as integrals over all field configurations weighted by $\exp(iS/\hbar)$. Loop diagrams generate ultraviolet divergences that renormalization manages by absorbing them into redefinitions of physical parameters. A renormalizable theory requires only finitely many such parameters.

Forces

- + LSZ reduction formula (in-out amplitudes from correlation functions)
- + Optical theorem (unitarity of S-matrix)
- + Feynman rules from path integral expansion
- + Crossing symmetry

Breaks on

- ✗ UV divergences: loop integrals diverge (load redistribution required = renormalization)
- ✗ Non-renormalizable theories: load grows without bound at each order (gravity)

Useful for

- Standard Model of particle physics
- Quantum electrodynamics
- Condensed matter (BCS superconductivity)
- Quantum computing (topological QFT)

73 Kolmogorov Probability Theory

Probability & Evidence

Andrei Kolmogorov (1933) *Grundbegriffe der Wahrscheinlichkeitsrechnung*

- V** Probability space $(\Omega, \mathcal{F}, P)$: sample space Ω , sigma-algebra $\mathcal{F}$ of events, probability measure $P: \mathcal{F} \rightarrow [0,1]$ with $P(\Omega)=1$ and sigma-additivity.
- G** Conditional probability $P(A|B)=P(A \cap B)/P(B)$. Independence: $P(A \cap B)=P(A)P(B)$. Expectation $E[X]=\int X dP$. Conditional expectation $E[X|\mathcal{F}_0]$: the $\mathcal{F}_0$ -measurable projection minimizing $E[(X-Y)^2]$ over $\mathcal{F}_0$ -measurable Y .
- Θ** $P(\Omega)=1$ and sigma-additivity: $P(\text{disjoint union } A_n)=\sum P(A_n)$. These two axioms are the complete specification.

PL INSIGHT

Kolmogorov's axiomatization identifies probability with measure on a sigma-algebra, making probability a special case of Lebesgue integration. The sigma-additivity axiom is the structural foundation: the probability of a countable disjoint union equals the sum of individual probabilities. Conditional expectation $E[X|\mathcal{F}_0]$ is defined not pointwise but as the unique $\mathcal{F}_0$ -measurable random variable minimizing $E[(X-Y)^2]$ -- a projection onto a closed subspace of $L^2(\Omega, \mathcal{F}, P)$. This Hilbert space structure underpins all of statistical inference and stochastic analysis. The entire theory follows from the axioms plus measure theory: Bayes' theorem, independence, law of large numbers, and central limit theorem all emerge from the arithmetic of P .

Forces

- + Sigma-additivity: $P(\text{disjoint union } A_n)=\sum P(A_n)$ for countable pairwise disjoint families
- + Monotonicity: $A \subseteq B$ implies $P(A) \leq P(B)$
- + Bayes' theorem: $P(A|B)=P(B|A)P(A)/P(B)$ -- an algebraic identity from the definition of conditional probability
- + Law of total probability: $P(A)=\sum P(A|B_i)P(B_i)$ for any partition $\{B_i\}$

Breaks on

- ✗ Zero-probability conditioning: $P(A|B)$ undefined when $P(B)=0$ (requires regular conditional distributions)
- ✗ Non-measurable events exist assuming AC -- probability not assignable to every subset of Ω
- ✗ Single probability measure cannot model all real-world uncertainty (motivating imprecise probability)

Useful for

- Mathematical statistics (estimation, hypothesis testing)
- Stochastic processes (Brownian motion, Markov chains)
- Bayesian analysis (posterior = prior $\times$ likelihood)
- Information theory (entropy as $-E[\log P]$)
- Mathematical finance (risk-neutral pricing, martingale theory)

Thomas Bayes (1763 posthumous); Laplace (1812) inverse probability; de Finetti (1937) subjective interpretation

- V** Prior $P(H)$ over hypotheses H ; likelihood function $P(E|H)$; posterior $P(H|E)$ via Bayes' theorem. Degrees of belief as coherent probability assignments over a hypothesis space.
- G** Bayes' theorem: $P(H|E) = P(E|H)P(H)/P(E)$. Sequential updating: each new observation multiplies the prior odds ratio by the likelihood ratio. Jeffrey conditionalization: $P'(A) = P(A|E)P'(E) + P(A|\neg E)P'(\neg E)$ for uncertain evidence.
- Ø** Coherence: a system of degrees of belief is coherent iff it satisfies Kolmogorov axioms; otherwise a Dutch book (guaranteed-loss betting strategy) exists. Cromwell's rule: never assign probability 0 or 1 to falsifiable hypotheses.

PL INSIGHT

Bayesian probability treats probability as a coherent degree of belief updated by evidence. The structural requirement is coherence: incoherent probability assignments allow Dutch books -- bets where the agent is guaranteed to lose regardless of outcomes. Bayes' theorem is the unique coherent updating rule given additive probability and conditionally independent evidence: $P(H|E) \propto P(E|H)P(H)$. The prior encodes beliefs before evidence; the likelihood encodes the discriminating power of the evidence for each hypothesis; the posterior integrates both. Unlike frequentist probability, Bayesian probability can assign probabilities to single events and to hypotheses themselves, making it the natural framework for scientific inference.

Forces

- + Dutch book coherence: violating the Kolmogorov axioms allows an adversary to construct bets guaranteeing a net loss regardless of outcome
- + Sequential consistency: updating on E_1 then E_2 equals jointly updating on (E_1, E_2) given independence
- + Bernstein-von Mises theorem: Bayesian posteriors converge to the frequentist MLE as data accumulates (with sufficient regularity)
- + Cromwell's rule: prior $P(H)=0$ can never be updated; $P(H)=1$ cannot be decreased -- evidence has no effect

Breaks on

- ✗ Prior sensitivity: different priors give different posteriors with limited data; no universal 'uninformative' prior
- ✗ Jeffreys prior solves the problem for certain exponential families but not universally
- ✗ Computational intractability: exact posterior requires integration over high-dimensional hypothesis space (motivating MCMC, variational inference)

Useful for

Machine learning (Bayesian classifiers, Gaussian processes, variational autoencoders)

Medical diagnosis (prior = disease prevalence, likelihood = test sensitivity/specificity)

Scientific inference (parameter estimation, model comparison)

A/B testing and experimental design

Spam filtering (naive Bayes classifier)

Arthur Dempster (1967) *upper and lower probabilities*; Glenn Shafer (1976) *A Mathematical Theory of Evidence*

- V** Frame of discernment Θ (finite set of mutually exclusive, exhaustive hypotheses). Basic probability assignment (BPA) $m: 2^\Theta \rightarrow [0,1]$ with $\sum m(A)=1$, $m(\emptyset)=0$. Belief $\text{Bel}(A)=\sum_{B \subseteq A} m(B)$. Plausibility $\text{Pl}(A)=1-\text{Bel}(\neg A)$.
- G** Dempster's combination rule: $(m_1 \oplus m_2)(A) = \sum_{B \cap C = A} m_1(B)m_2(C) / K$ where $K = 1 - \sum_{B \cap C = \emptyset} m_1(B)m_2(C)$ normalizes away contradictions. $K=0$: sources are completely contradictory.
- Θ** Belief-plausibility interval $[\text{Bel}(A), \text{Pl}(A)]$: lower and upper probability bounds on A. $\text{Pl}(A) - \text{Bel}(A)$ measures epistemic uncertainty. Consonant BPA (nested focal sets): $\text{Bel} = \text{Pl}$ (no uncertainty, equivalent to classical probability).

PL INSIGHT

Dempster-Shafer theory separately represents ignorance and uncertainty. A BPA assigns mass to sets of hypotheses (focal elements) when evidence does not discriminate among them: $m(\{H1, H2\})=0.7$ means the evidence supports one of H1 or H2 but cannot determine which. Belief $\text{Bel}(A)$ is the total mass committed exclusively to subsets of A -- the lower probability bound. Plausibility $\text{Pl}(A)$ includes all mass not excluded from A -- the upper bound. The interval $[\text{Bel}(A), \text{Pl}(A)]$ is the evidential statement: probability lies somewhere in this interval. Complete ignorance: $m(\Theta)=1$ gives $[0,1]$ for every non-trivial A. This allows expressing genuine ignorance as distinct from uniform uncertainty.

Forces

- + $\text{Bel}(A) \leq \text{Pl}(A)$ always (belief is lower bound, plausibility is upper bound on A's probability)
- + $\text{Bel}(\Theta)=1$ and $\text{Pl}(\Theta)=1$ (total belief in full hypothesis space)
- + Vacuous BPA $m(\Theta)=1$: $\text{Bel}(A)=0$, $\text{Pl}(A)=1$ for all proper subsets A -- represents complete ignorance
- + Classical probability is recovered when all focal elements are singletons: $\text{Bel} = \text{Pl} = P$

Breaks on

- ✗ Zadeh's paradox: combining two near-certain sources each supporting different single hypotheses can give high confidence in a third hypothesis neither source supports
- ✗ Dempster's rule assumes independence of sources; combining dependent sources can give counterintuitive results
- ✗ Not associative in general when conditioning is involved

Useful for

- Expert systems and knowledge engineering
- Multi-sensor fusion (robotics, military target recognition)
- Fault diagnosis in engineering systems
- Decision making under genuine ignorance
- Forensic evidence combination (court reasoning)

Zadeh (1978) *Fuzzy Sets and Systems*; Dubois & Prade (1988) *Possibility Theory: An Approach to Computerized Processing of Uncertainty*

- V** Possibility distribution $\pi: \Omega \rightarrow [0,1]$ with $\sup_{\omega} \pi(\omega) = 1$. Possibility measure $\Pi(A) = \sup_{x \in A} \pi(x)$. Necessity measure $N(A) = 1 - \Pi(\neg A) = \inf_{x \notin A} (1 - \pi(x))$.
- G** $\Pi(A \cup B) = \max(\Pi(A), \Pi(B))$ (max-decomposable). $N(A \cap B) = \min(N(A), N(B))$. Duality: $N(A) = 1 - \Pi(\neg A)$. Possibility conditioning: $\Pi(A|B) = \Pi(A \cap B) / \Pi(B)$.
- 0** Normalization: $\sup_{x \in \Omega} \pi(x) = 1$ (at least one outcome is fully possible). Complete ignorance: $\pi \equiv 1$ everywhere. Certainty: $\pi(x_0) = 1$, $\pi(x) = 0$ for $x \neq x_0$.

PL INSIGHT

Possibility theory replaces probability's additivity with max-decomposability. $\Pi(A \cup B) = \max(\Pi(A), \Pi(B))$ means the possibility of a union is the maximum of the parts, not their sum. This allows $\Pi(A) = \Pi(\neg A) = 1$ simultaneously: both A and $\neg A$ are fully possible when no evidence discriminates between them -- complete ignorance. Necessity $N(A) = 1 - \Pi(\neg A)$ measures certainty: $N(A) = 1$ means $\neg A$ is impossible. $N(A) = 0$ means $\neg A$ is at least partially possible. The pair (Π, N) gives a two-sided characterization of what is known: possibility is an upper bound (what could be), necessity is a lower bound (what must be). This models linguistic vagueness and imprecise information more naturally than forcing a single probability distribution.

Forces

- + $\Pi(\Omega) = 1$, $\Pi(\emptyset) = 0$ (possibility of the full space = 1, of empty set = 0)
- + $\Pi(A \cup B) = \max(\Pi(A), \Pi(B))$ -- max-decomposable, not additive
- + $N(A) + \Pi(\neg A) = 1$ (duality between necessity and possibility)
- + Vacuous distribution: $\pi = 1$ everywhere gives $N(A) = 0$, $\Pi(A) = 1$ for all non-trivial A -- complete ignorance

Breaks on

- ✗ Non-additivity by design: $\Pi(A) + \Pi(\neg A) \geq 1$ always, so both A and $\neg A$ can simultaneously have possibility 1 (complete ignorance)
- ✗ Possibility theory does not obey probability's additivity -- it is a max-based, not sum-based, calculus
- ✗ No unique possibility distribution corresponds to a given probability distribution (probability-possibility transformations are not unique)

Useful for

- Fuzzy control systems (uncertainty propagation through controllers)
- Expert systems (imprecise linguistic knowledge)
- Natural language semantics (vague terms like 'tall', 'old', 'likely')
- Risk analysis under severe information gaps
- AI planning under incomplete information

Walley (1991) *Statistical Reasoning with Imprecise Probabilities*; Weichselberger (2001); Kuznetsov (1991)

- V** Credal set K : convex set of probability measures on $(\Omega, \mathcal{I})$. Lower probability $P_*(A) = \min_{P \in K} P(A)$. Upper probability $P^*(A) = \max_{P \in K} P(A)$. Lower expectation $E_*(X) = \min_{P \in K} E_P[X]$.
- G** Natural extension: coherently extend lower prevision to all bounded random variables. Robust Bayes: update each $P \in K$ by Bayes' rule to get posterior credal set. Marginal extension for independent variables.
- Ø** Credal set K represents available information. $K = \text{singleton}$: classical probability. $K = \text{all measures on } \Omega$: vacuous (complete ignorance). Posterior K shrinks as evidence accumulates.

PL INSIGHT

Imprecise probability replaces a single probability measure with a convex set of compatible measures (credal set) to represent genuine uncertainty about which probability model is correct. The lower probability $P_*(A)$ is the minimum probability of A across all measures in K ; $P^*(A)$ is the maximum. Classical probability is the special case $K = \{P\}$ (singleton). Complete ignorance is $K = \text{all measures on } \Omega$, giving $P_*(A) = 0$, $P^*(A) = 1$ for all non-trivial A . Walley's coherence condition generalizes the Dutch-book argument: a system of lower previsions is coherent iff no adversary can extract a guaranteed gain by combining accepted bets. This is the broadest generalization of probability theory that still supports rational decision-making.

Forces

- + $P_*(A) \leq P^*(A)$ always; $P_*(A) + P^*(\neg A) = 1$ (conjugacy)
- + Coherence: no Dutch book against the lower prevision -- the behavioral interpretation of lower probabilities
- + Posterior credal set obtained by Bayes-updating each $P \in K$: gives interval-valued posteriors
- + Updating by evidence shrinks the credal set (information reduces imprecision) under mild conditions

Breaks on

- ✗ Decision theory becomes set-valued: multiple admissible acts may exist when using imprecise probabilities (E-admissibility, maximality, gamma-maximin)
- ✗ Dilation: conditioning on an event can sometimes increase imprecision rather than reduce it
- ✗ Computing lower/upper probabilities over credal sets is often NP-hard in the number of extreme points

Useful for

- Robust statistics (conclusions valid across a class of models)
- Risk analysis under deep uncertainty (financial systemic risk, climate change)
- Artificial intelligence (credal networks)
- Sensitivity analysis (how much do conclusions depend on model assumptions?)
- Medical decision making under genuine ignorance

distinction.py (2026); sample-complexity and concentration bounds (Chernoff, Hoeffding); Shannon (1948) coding theory

- V** Pairs of candidate values to be told apart against noise. A point is 'two hypotheses separated by a gap', and the question is how much evidence it takes to designate which one is true at confidence $1 - \delta$.
- G** Repeated noisy probing: sequential sampling that averages down variance, and error-correcting repetition that trades redundancy for reliability. Each probe is one application of the gradient; the load is the count of probes (looks, code bits, iterations, ticks).
- θ** The confidence threshold $1 - \delta$. A distinction coheres only once accumulated evidence separates the two values with error at most δ ; below that it is undecided.

PL INSIGHT

This carrier prices what Chapter 6 said was never free: the cost of maintaining a distinction. The wrong first move is to price it by the raw distance between the two values, which makes near-boundary calls look cheap. The correction swaps distance for the number of noisy looks needed to be sure, which scales as $1/\text{gap}^2$ -- so the distinction at the boundary is not cheap but unbounded, matching the intuition that the medical test at threshold or the speed camera at the limit is where cost explodes. Measured across four systems, two regimes appear: a variance-tax of gap^{-2} where noise must be averaged down, and a lighter rate-tax of gap^{-1} where a process converges geometrically. The deepest identity is that error correction and learning are the same loop: a seeker with no prior knowledge probes a boundary, feels coherence break, detects it, and retreats just inside the edge -- rediscovering a solver's optimal relaxation parameter from below. The whole thing hashes its own source and grades every claim by tier so that no result is dressed above its evidence.

Forces

- + The Boundary Law: $D = 8 \sigma^2 \ln(1/\delta) / \text{gap}^2$ -- the cost to tell two values apart
- + Cost diverges as $\text{gap} \rightarrow 0$: the near-boundary distinction is the most expensive, not the cheapest (correcting the starter formula $|a-b|/\text{range}$)
- + Variance-tax (gap^{-2}): systems that average down noise (testing, coding) pay an inverse-square bill; halving the gap quadruples the cost
- + Rate-tax (gap^{-1}): systems that converge geometrically (solver, cascade) pay only an inverse-linear bill
- + Error correction is learning: probe, decohere, detect, retreat is one loop -- $P/G \rightarrow Q$ with the error as load and the retreat as reconfiguration

Breaks on

- ✗ The measured exponents are per-system, in each system's own load unit -- universality across all carriers is NOT claimed (STIPULATED)
- ✗ Noise is modelled Gaussian (or Bernoulli where stated); heavy-tailed or adversarial noise can change the exponent
- ✗ It is measure-and-bisect with a divergence guard, not novel optimisation -- the contribution is the frame, not the algorithm (UNPAID)

Useful for

Sample-complexity analysis (how many trials to distinguish two hypotheses)

Error-correcting code design (repetition and redundancy versus channel noise)

Sequential testing and A/B experiment sizing near a decision boundary Active learning and bandit exploration (probe, detect, retreat)

Pricing measurement and classification cost in reified binary decisions (Chapter 6's distinction debt, made quantitative)

78 Copenhagen Interpretation

Measurement Interpretations

Niels Bohr (1927) complementarity; Heisenberg (1927) uncertainty; formalized in Bohr-Einstein debates (1927-1935)

- V** Quantum state $|\psi\rangle$ in Hilbert space H (pre-measurement). Classical measurement outcome λ_i (eigenvalue) post-measurement. No intermediate states defined; no description of particle between measurements.
- G** Pre-measurement: Schrodinger evolution $U(t)|\psi\rangle$. Measurement: projective collapse $|\psi\rangle \rightarrow |\lambda_i\rangle$ (eigenstate of observable). Probability: $|\langle \lambda_i | \psi \rangle|^2$ (Born rule). No mechanism for collapse is specified.
- Ø** The Heisenberg cut: an unspecified boundary between quantum (superposition, wave-like) and classical (definite, particle-like) regimes. The measuring apparatus must be treated classically.

PL INSIGHT

Copenhagen treats quantum mechanics as a theory about measurement outcomes rather than observer-independent reality. The quantum state $|\psi\rangle$ is an epistemic tool for computing probabilities; it does not describe the particle's definite properties between measurements. Measurement irreversibly actualizes one outcome from the quantum superposition; no mechanism for this collapse is specified -- it is taken as a primitive transition. Bohr's complementarity principle asserts that wave and particle descriptions are mutually exclusive but jointly necessary: the experimental context (which observable is measured) determines which classical concept applies. The theory works empirically with extraordinary precision but deliberately leaves the ontological question of what happens between measurements unanswered.

Forces

- + Born rule: probability of outcome λ_i is $|\langle \lambda_i | \psi \rangle|^2$
- + Complementarity: conjugate observables (position and momentum, energy and time) cannot both have definite values simultaneously
- + Measurement irreversibly actualizes one outcome; the pre-measurement superposition cannot be recovered
- + Wave-particle duality: the experimental arrangement determines which classical description (wave or particle) is applicable

Breaks on

- × No precise specification of where the Heisenberg cut lies (the 'measurement problem')
- × Collapse is instantaneous -- formally non-relativistic in appearance, though no information propagates faster than light
- × EPR correlations: entangled pairs violate Bell inequalities, appearing to require nonlocal influence (though no signal is transmitted)
- × Schrodinger's cat paradox: macroscopic superpositions are physically undefined in Copenhagen

Useful for

- Standard quantum mechanics textbooks and courses
- Practical quantum computing (state preparation and measurement)
- Engineering quantum systems (MRI, laser design, electron microscopy)
- Spectroscopy and quantum optics
- The working physicist's default operational framework

Hugh Everett III (1957) relative state formulation; Wheeler (1957) supervisor; DeWitt (1970) popularized 'many worlds' terminology

- V** Universal wavefunction Ψ in $H = H_{\text{system}} \otimes H_{\text{apparatus}} \otimes H_{\text{environment}}$. No collapse postulate. Branching structure defined post-hoc by decoherence selecting a preferred pointer basis.
- G** Unitary evolution only: $\Psi(t) = U(t)\Psi(0)$ for all t , including during measurement. Measurement = entanglement: $\Psi \rightarrow \sum_i c_i |\text{system}_i\rangle \otimes |\text{apparatus}_i\rangle \otimes |\text{environment}_i\rangle$. Decoherence: environment rapidly diagonalizes the reduced density matrix in the pointer basis.
- Θ** Decoherence time τ_D : after τ_D the off-diagonal terms of ρ_{reduced} are effectively zero; branches do not interfere on accessible timescales.

PL INSIGHT

Many-Worlds eliminates the collapse postulate entirely: the universal wavefunction Ψ evolves unitarily at all times, including during measurement. Measurement produces entanglement between the system, apparatus, and environment, causing Ψ to become a superposition of product terms each containing a definite outcome. Decoherence prevents interference between these terms on any accessible timescale, making the branches effectively separate. Each branch is an observer's 'relative state' -- their classical world in which one definite outcome occurred. The apparent randomness of quantum mechanics emerges from the observer's uncertainty about which branch they inhabit; the fundamental evolution is deterministic.

Forces

- + Unitarity: Ψ evolves by Schrodinger equation at all times without exception
- + All measurement outcomes occur in branches: each term in the post-measurement superposition is a 'relative state' (Everett) or 'branch' (DeWitt)
- + Born rule must be derived, not postulated: Deutsch-Wallace decision-theoretic argument or Zurek's enviance gives $P(\text{outcome}_i) = |c_i|^2$
- + Decoherence enforces an approximate branch structure in the pointer basis selected by environmental coupling

Breaks on

- ✗ Ontological extravagance: all outcomes occur in parallel branches -- the total 'amount of existence' is distributed but not reduced
- ✗ Born rule derivation is contested: the decision-theoretic argument requires additional rationality axioms that some find circular
- ✗ The 'branch' definition is approximate: decoherence selects a preferred basis approximately but not exactly

Useful for

Quantum gravity and cosmology (no external observer for the universe)

Theoretical motivation for quantum computing (parallel computation in branches)

Foundations of quantum information (understanding entanglement without collapse)

Philosophy of probability (what does 'probability' mean in a deterministic many-worlds theory?)

Conceptual basis for decoherence theory

Louis de Broglie (1927) pilot wave; David Bohm (1952) A Suggested Interpretation of Quantum Theory in Terms of Hidden Variables

- V** Configuration space $Q_N = \mathbb{R}^{3N}$ for N particles with wavefunction $\Psi: Q_N \rightarrow \mathbb{C}$ (always defined) PLUS actual particle positions $Q = (Q_1, \dots, Q_N) \in Q_N$ (hidden variables, always definite).
- G** Schrodinger equation for $\Psi(q, t)$: standard, unmodified. Guiding equation: $dQ_k/dt = (\hbar/m_k) \text{Im}(\nabla_k \Psi / \Psi)$ evaluated at actual positions Q . Ψ guides particles; particles do not influence Ψ .
- Θ** Quantum equilibrium: initial distribution $\rho_0(q) = |\Psi_0(q)|^2$. Equivariance: if $\rho_0 = |\Psi_0|^2$ then $\rho_t = |\Psi_t|^2$ for all t -- the Born rule distribution is preserved by the dynamics.

PL INSIGHT

Bohm's theory restores determinism by adding actual particle positions Q to the wavefunction Ψ . Ψ guides the particles via the guiding equation but is not itself influenced by the particles. The wavefunction is a 'pilot wave' physically guiding particle trajectories. Born rule probabilities are not postulated but derive from quantum equilibrium: if the initial distribution of positions matches $|\Psi_0|^2$, the equivariance theorem guarantees it remains so. What Copenhagen calls 'collapse' is in Bohm's theory a selection: the actual particle position determines which branch of Ψ is effective for future guidance, while other branches persist as real but empty. The theory is empirically identical to standard QM but ontologically explicit: both Ψ and Q are always real.

Forces

- + Born rule: given quantum equilibrium ($\rho_0 = |\Psi_0|^2$), measurement outcome probabilities agree with standard QM for all time
- + Determinism: given $Q(0)$ and $\Psi(0)$, the entire trajectory $Q(t)$ is uniquely determined by the guiding equation
- + No collapse: Ψ evolves by Schrodinger equation always; apparent collapse is selection by the actual particle trajectory
- + Equivariance theorem: the quantum equilibrium distribution is preserved -- $\rho_t = |\Psi_t|^2$ for all t

Breaks on

- ✗ Non-locality: the guiding equation for particle k depends on all other particle positions instantaneously -- required by Bell's theorem but conflicts with special relativity
- ✗ Preferred foliation: the instantaneous nonlocal guidance requires a preferred reference frame (non-relativistic in character)
- ✗ Empty branches: wavefunction branches where the actual particle is absent persist as physically real but 'empty' waves

Useful for

- Analysis of EPR and Bell inequality violations (explicit nonlocal mechanism)
- Foundations of quantum mechanics (makes ontology explicit)
- Quantum chaos (Bohmian trajectories provide intuition)
- Extensions to quantum field theory (field beable versions)
- Pedagogical tool for understanding quantum trajectories

Griffiths (1984); Gell-Mann & Hartle (1990, 1993) decoherent histories; Omnès (1992) logical interpretation

- V** History: sequence of projection operators at times $t_1 < \dots < t_n$: $\alpha = (P^{\alpha_1}_{t_1}, \dots, P^{\alpha_n}_{t_n})$. Homogeneous history set: exhaustive collection of exclusive histories. Class operator $C_\alpha = P^{\alpha_n}_{t_n} U(t_n, t_{n-1}) \dots P^{\alpha_1}_{t_1}$.
- G** Decoherence functional: $D(\alpha, \alpha') = \text{Tr}(C_\alpha P_0 C^\dagger_{\alpha'})$. Consistency condition: $\text{Re } D(\alpha, \alpha') = 0$ for $\alpha \neq \alpha'$. Probability within consistent family: $P(\alpha) = D(\alpha, \alpha) = \text{Tr}(C_\alpha P_0 C^\dagger_\alpha)$.
- Ø** Consistency (decoherence): $D(\alpha, \alpha') = 0$ for all $\alpha \neq \alpha'$ within a family. Only consistent frameworks permit probability assignments and classical reasoning.

PL INSIGHT

Consistent histories assigns probabilities to sequences of events (histories) without requiring observers or measurement apparatus. A set of histories is consistent when off-diagonal decoherence functional elements vanish: the histories do not interfere with each other, so classical probability rules apply within the framework. The theory is explicitly contextual: different consistent frameworks ('realms') are equally valid descriptions of the same physics, and logical inferences can only be drawn within a single framework. This avoids the measurement problem by recognizing that 'what happened' is framework-relative: the same physical evolution can be described by incompatible but internally consistent sets of histories.

Forces

- + Within a consistent family: probabilities sum to 1 and satisfy classical Kolmogorov axioms
- + Born rule recovered: $P(\alpha) = |C_\alpha \psi|^2$ for pure initial state $|\psi\rangle$ and orthogonal projections
- + Standard QM is a special case: single-time measurements trivially form a consistent family
- + Classical physics is recovered: sufficiently coarse-grained, macroscopic histories are always approximately consistent

Breaks on

- × No unique framework: the same physical situation has many mutually incompatible consistent families -- no rule selects which to use
- × Contextuality: propositions valid in one framework may be meaningless or false in another
- × Combining statements from different consistent frameworks leads to contradictions (Frauchiger-Renner type issues)

Useful for

- Quantum cosmology (reasoning about the universe without an external observer)
- Closed quantum systems (no measurement apparatus required)
- Foundational analysis of decoherence and the quantum-to-classical transition
- Quantum gravity (reasoning about space-time histories)
- Philosophical analysis of quantum logic and counterfactual reasoning

Ghirardi, Rimini & Weber (1986) GRW spontaneous localization; Penrose (1989, 1994) objective reduction (OR)

- V** Quantum state $|\Psi\rangle$ in Hilbert space H . Spontaneous stochastic collapse events occur at rate λ per particle (GRW). Penrose OR: collapse triggered when superposition involves gravitationally distinguishable mass distributions.
- G** Normal Schrodinger evolution between collapses. GRW collapse: randomly selected particle k undergoes localization $|\Psi\rangle \rightarrow L_{\{k,x\}}|\Psi\rangle / \|L_{\{k,x\}}|\Psi\rangle\|$ where $L_{\{k,x\}}$ is Gaussian centered at random x drawn from probability $\|L_{\{k,x\}}|\Psi\rangle\|^2$. Penrose: collapse timescale $T \sim \hbar / (\Delta E)$ where ΔE = gravitational self-energy of mass distribution difference.
- Θ** GRW parameters: $\lambda \sim 10^{-16} \text{ s}^{-1}$ per particle (collapse rate), $\sigma \sim 10^{-7} \text{ m}$ (localization width). These are chosen so single particles rarely collapse while macroscopic objects collapse almost instantly.

PL INSIGHT

Objective collapse theories modify the Schrodinger equation by adding stochastic nonlinear collapse terms acting on all particles at rate λ per particle. The collapse rate per particle is tiny ($\lambda \sim 10^{-16} \text{ s}^{-1}$), so single particles maintain quantum coherence over long times. But for a macroscopic object with $N \sim 10^{23}$ particles, the aggregate rate $N\lambda \sim 10^7 \text{ s}^{-1}$ is enormous -- any attempt to maintain a macroscopic superposition is immediately destroyed. This naturally explains the quantum-to-classical transition without invoking observers: the stochastic collapse mechanism IS the physical process that actualizes definite outcomes. Penrose proposes that the collapse is triggered by quantum gravity -- superpositions of macroscopically different mass distributions are gravitationally unstable.

Forces

- + Born rule: GRW collapse probabilities reproduce $|\Psi|^2$ for all practical purposes for systems with many particles
- + Macro-objectivity: for $N \sim 10^{23}$ particles, aggregate collapse rate $N\lambda \sim 10^7 \text{ s}^{-1}$ -- macroscopic superpositions collapse almost instantly
- + Microscopic coherence: single particles have collapse rate $\lambda \sim 10^{-16} \text{ s}^{-1}$ -- quantum behavior preserved on particle scales
- + Schrodinger equation is modified: GRW adds stochastic nonlinear terms that dominate only for large N

Breaks on

- × Small energy non-conservation at each collapse event: the Gaussian localization increases the expected energy slightly
- × No complete relativistic extension of GRW: relativistic versions (CSL) exist but require additional structure
- × Penrose OR requires quantum gravity and is not yet a complete theory
- × No direct experimental evidence distinguishing objective collapse from standard QM at currently accessible scales

Useful for

Experimental tests at mesoscopic scales (levitated optomechanics, massive superposition experiments)

Quantum biology (is decoherence in photosynthesis explained by GRW-scale collapses?)

Philosophical resolution of the measurement problem (no observers needed)

Phenomenology of quantum gravity at accessible energies

Motivating future experiments to distinguish interpretations

Carlo Rovelli (1996) *Relational Quantum Mechanics*, *International Journal of Theoretical Physics* 35, 1637-1678

- V** Quantum states are relative to observers: $|\Psi\rangle_{A/O}$ (state of system A as described by observer O). No observer-independent quantum state. Fundamental ontology: interaction events between systems are the physical facts.
- G** Interactions actualize relative facts: when O interacts with A, O acquires a definite value for some observable of A, relative to O. A third observer O' can describe the O-A interaction using a combined quantum state. State descriptions are indexical to the describing system.
- Ø** Interaction events: a quantum event (definite outcome) occurs at each physical interaction between two quantum systems. The value is real relative to the observing system but not absolute.

PL INSIGHT

Relational QM applies the lesson of special relativity to quantum states: just as simultaneity is relative to an observer's frame, the quantum state of a system is defined only relative to another system. A quantum event (a definite measurement outcome) is real relative to the observing system but not absolute. When O measures A, A's state relative to O collapses to a definite value. A third observer O' who has not interacted with either can still describe O+A using a quantum superposition -- this is not a contradiction but a reflection of the fact that different observers have different interaction histories. The theory dissolves the measurement problem: 'collapse' is not a physical process but the update of O's relational description upon interaction.

Forces

- + Each observer's account is internally consistent: within O's description, standard QM applies without contradiction
- + Born rule applies within each observer's description: probabilities match standard predictions
- + No contradiction in any observable prediction: different observers' descriptions cannot be compared in a way that yields contradictory predictions
- + Wigner's friend and similar puzzles dissolve: each agent's state description is valid relative to them; their descriptions are not required to agree

Breaks on

- × No observer-independent quantum state: the ontological revision required is that 'the quantum state of A' is not an absolute fact
- × Cross-perspective combination of relational facts requires care (Wigner-friend scenarios show that combining different observers' descriptions requires knowing who has interacted with whom)
- × No explanation offered for why different observers' measurements are correlated: correlations are taken as primitive physical facts

Useful for

- Loop quantum gravity (relational frameworks are natural in background-independent theories)
- Quantum information (understanding correlations without external reference frames)
- Foundational analysis of Bell inequalities (nonlocality as a relational, not absolute, phenomenon)
- Philosophy of physics (observer-dependent ontology) Quantum reference frames